

Box Robot Base

version 1.0, rev 16 28 December 2025

This is a classic box robot design intended to provide two forward facing attachments or a combination of forward and sidefacing attachments. We usually configure the left attachment motor for our worm drive attachment and the right attachment motor, rotated at 90 degrees, is usually for a geared lifting arm. We also use the right motor to drive a forward facing axle to turn objects on the FLL board (like the gears in Masterpiece Mission 12 rotating the chicken or Unearthed Mission 11/Angler Artifact).

We recommend using four large Spike Prime motors. If you only have two large motors use them for the attachments and use the medium Spike Prime motors for the wheels. Using the medium motors will require adjustments to the design of the front assembly section.

When assembling the robot care needs to be taken on how the sensor and motor cables need to be routed to the hub prior to joining the front and rear sections. Secure the cables using multiple wire connectors (49283) in the rear compartment containing the hub.

It is loosely based on:

<https://www.youtube.com/watch?v=PCMEvf4p2PI&t=2s>

You might want to use a
4L axle instead of the 5L.

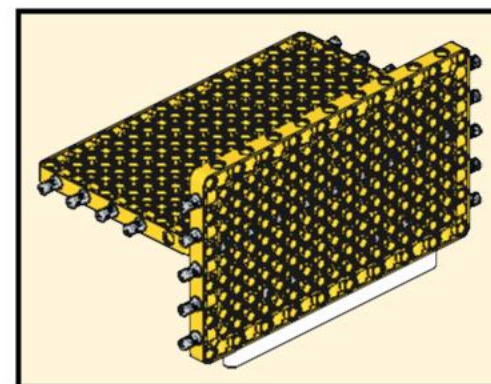
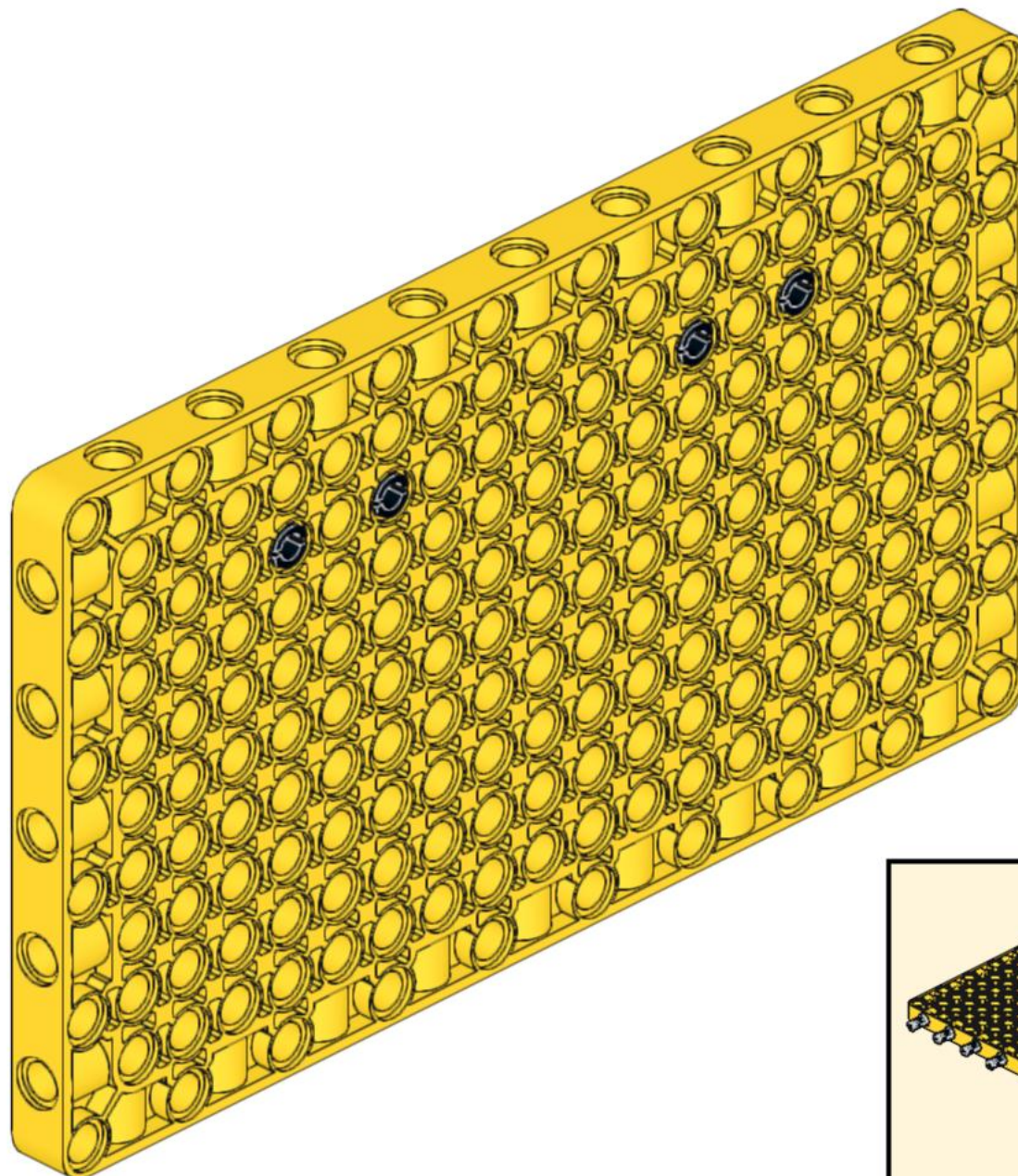
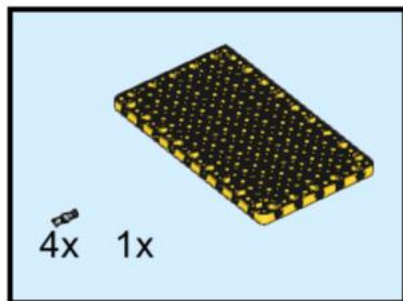
Front Assembly

Be careful on how attachment motors are placed on top of the yellow plate covering the driving motors.

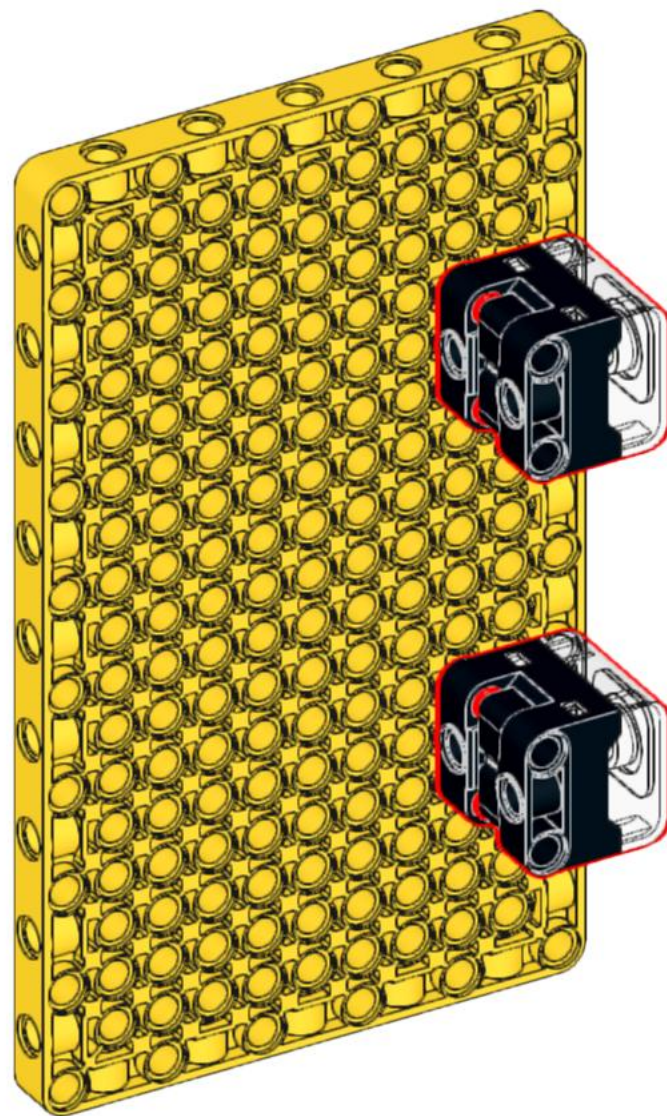
The primary mechanism the drive motors are secured is via attachment to this plate.

You need to optimize the number of pins (8) from the drive and attachment motors that are connected to the top and bottom of the plates.

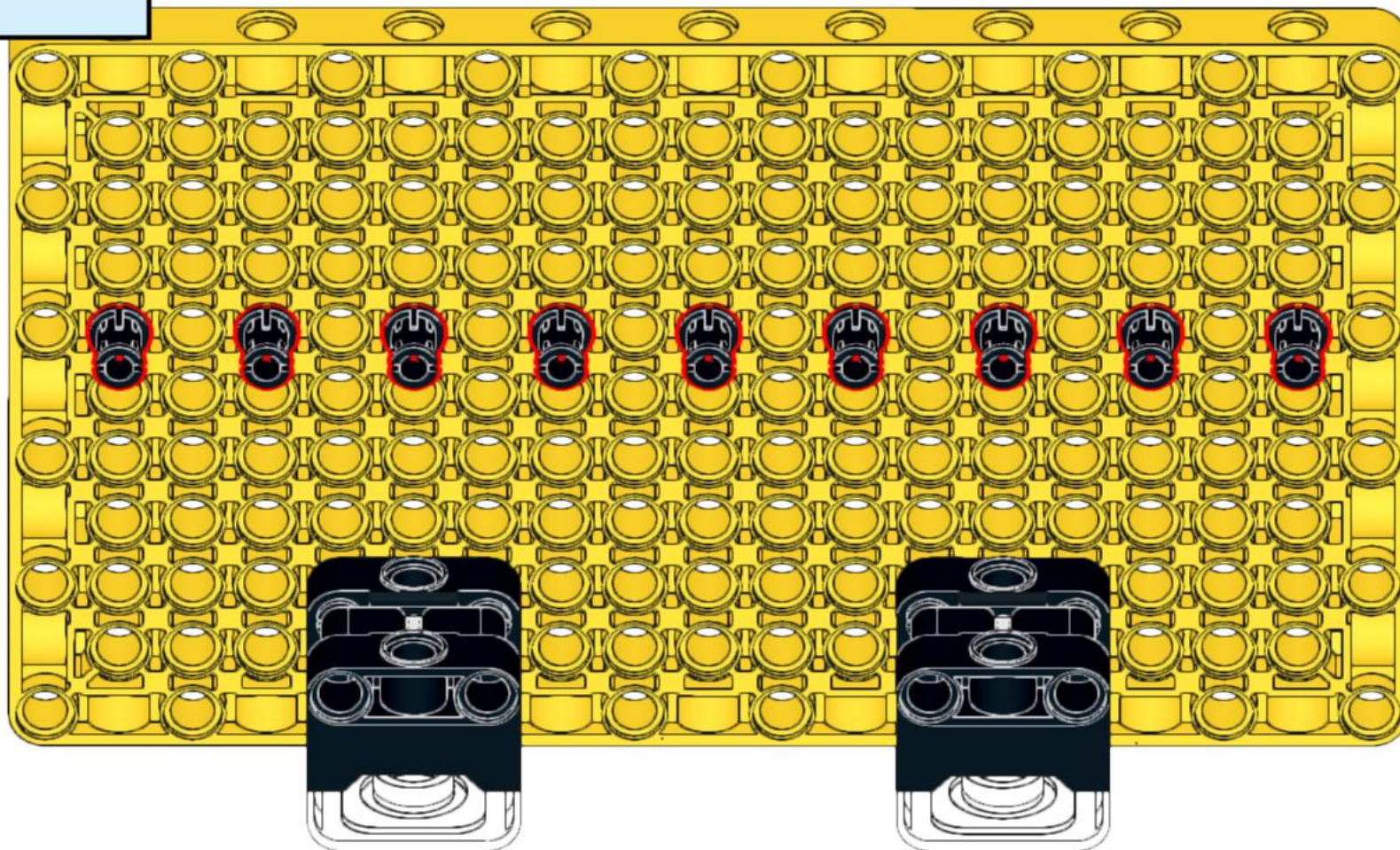
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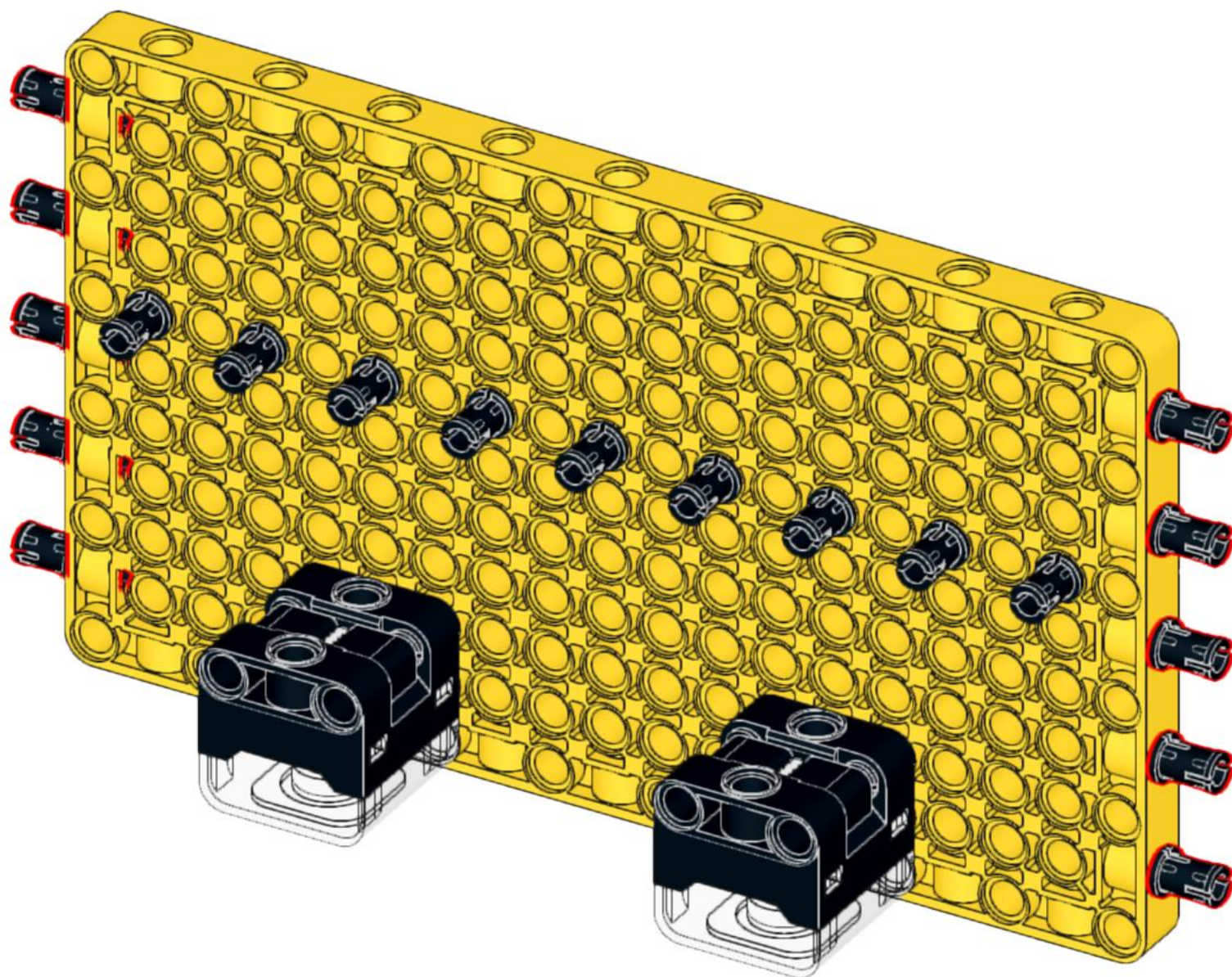
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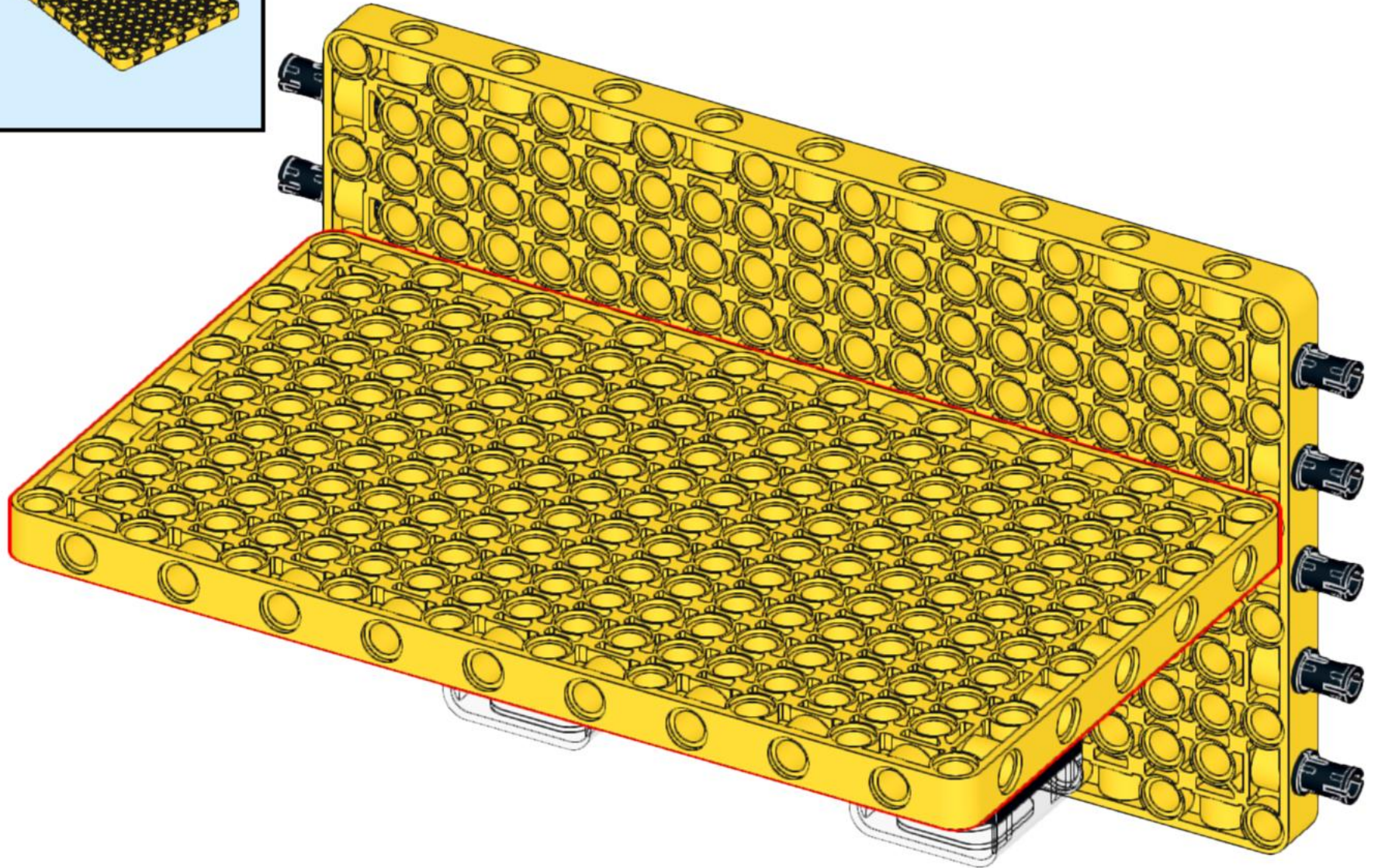
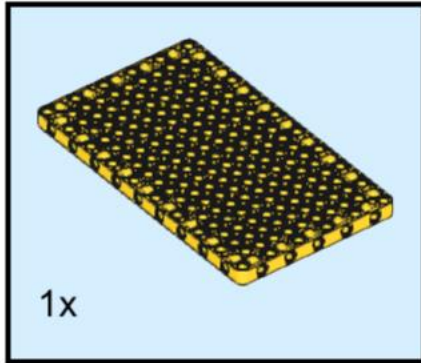
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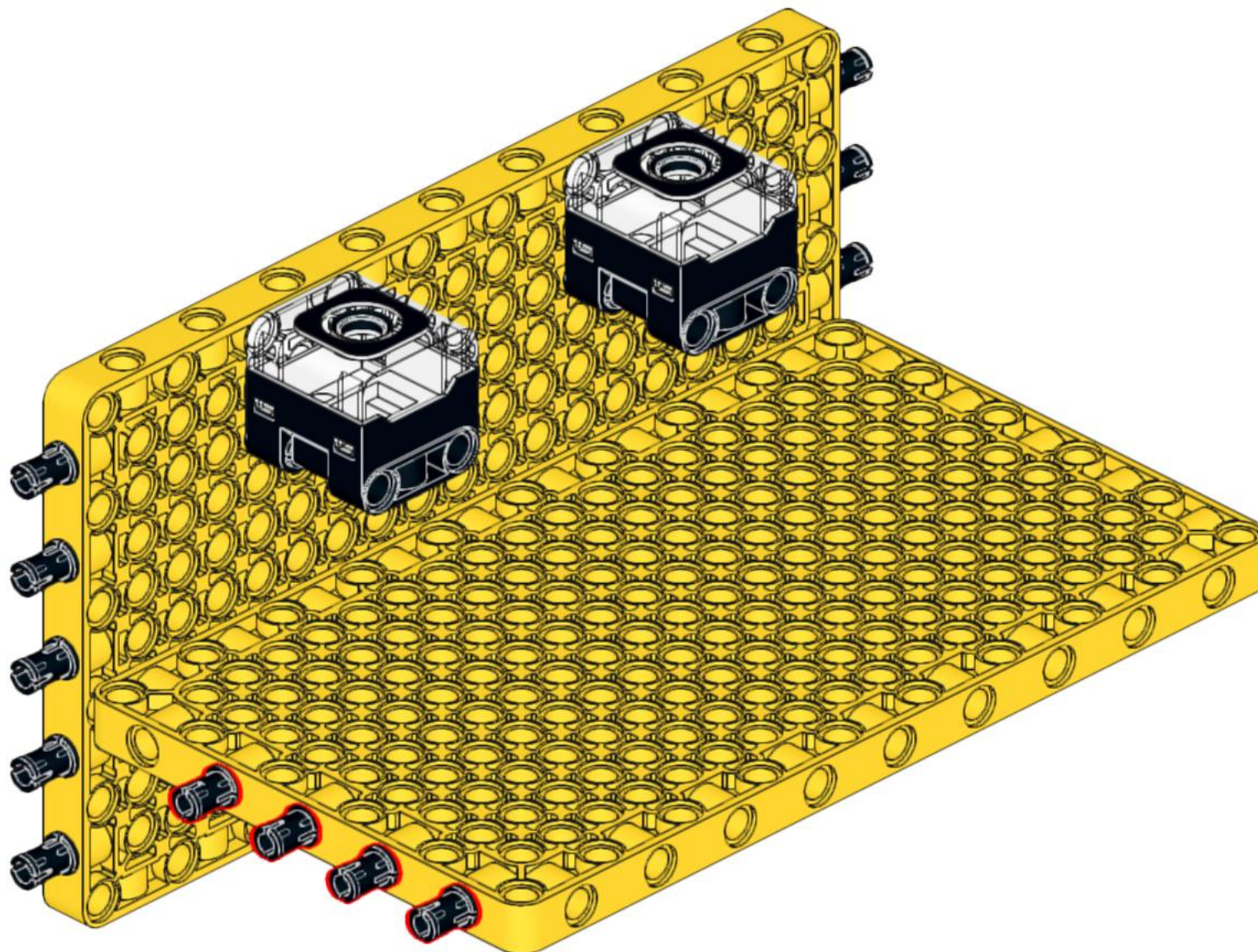
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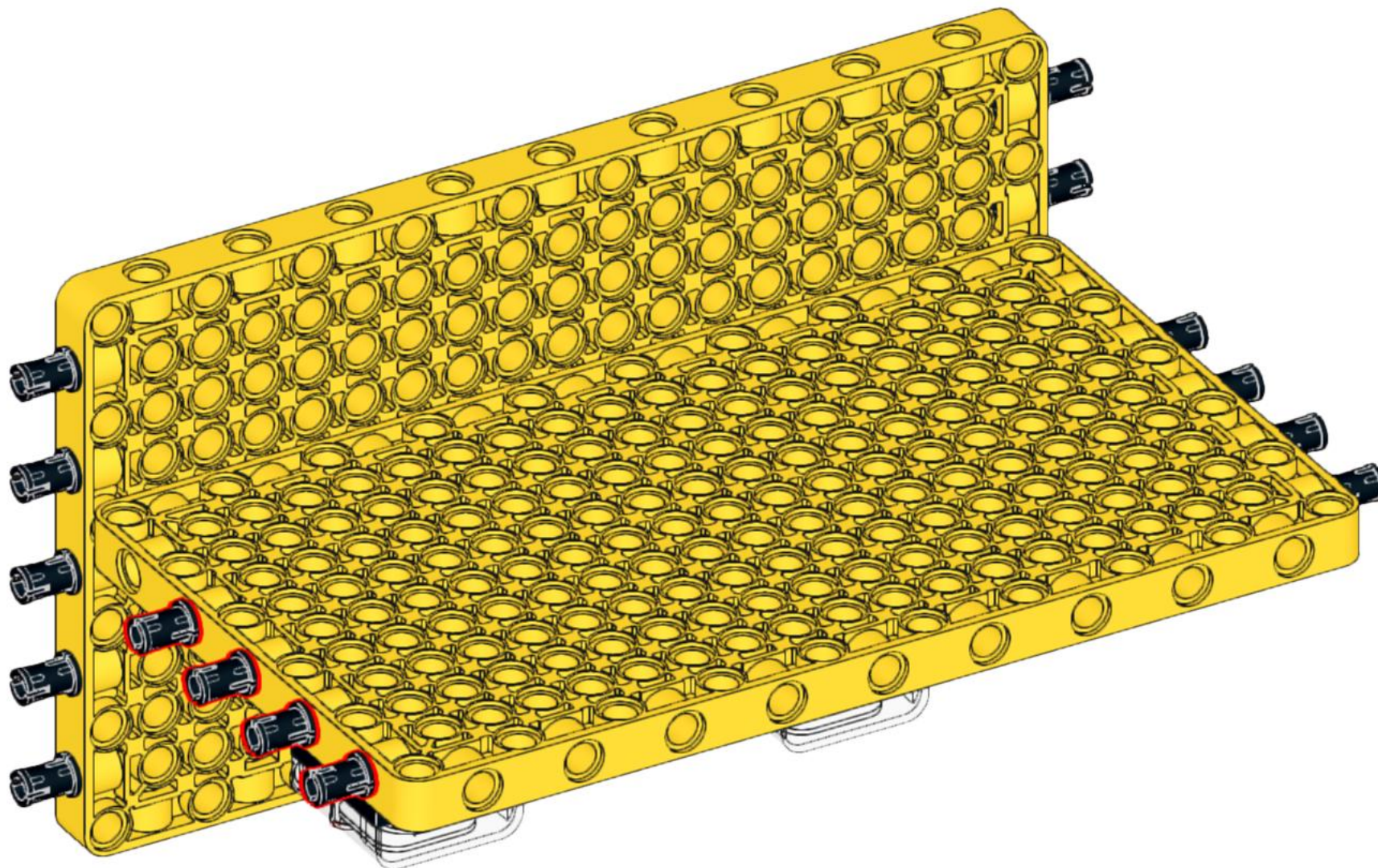
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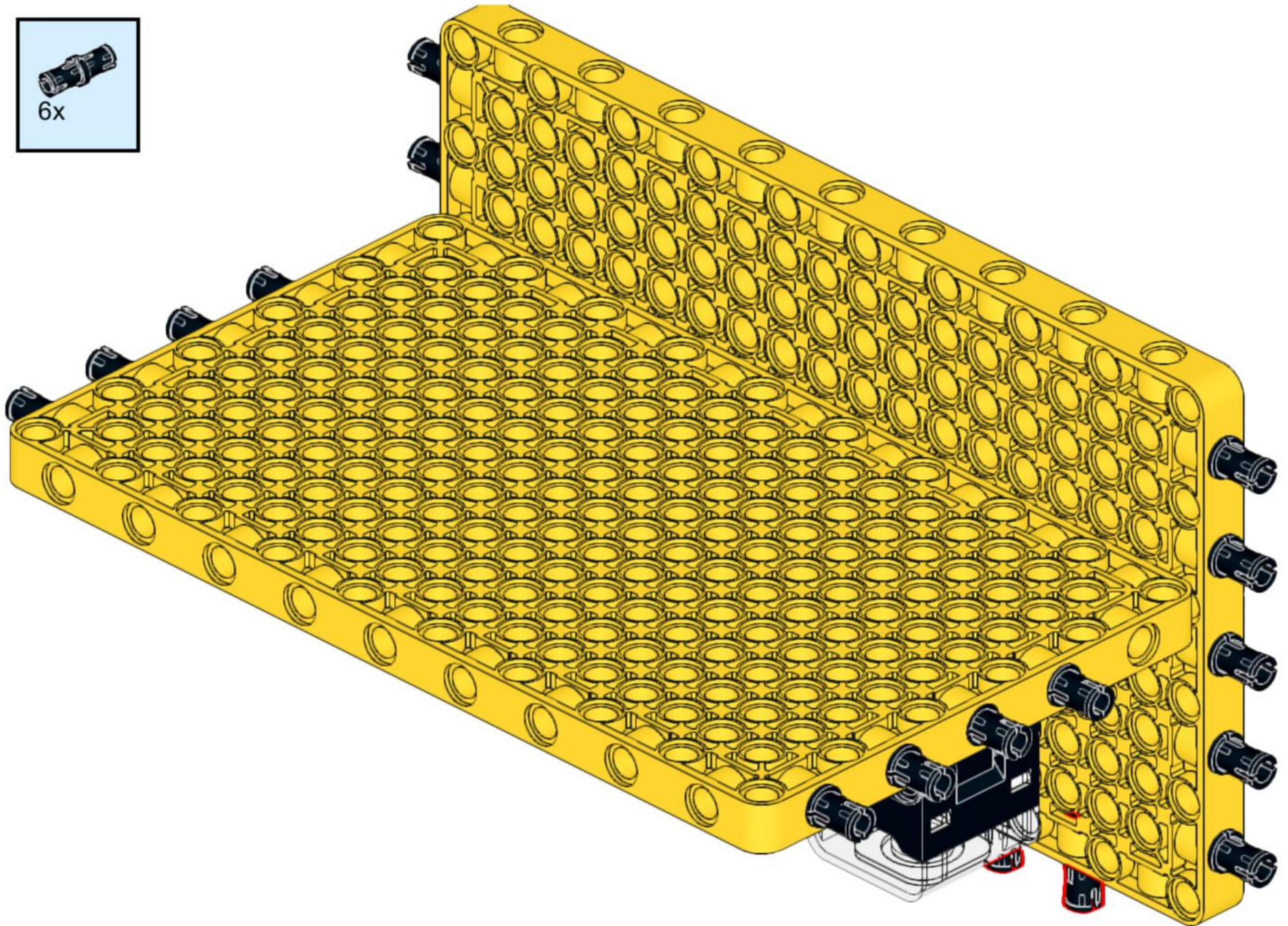
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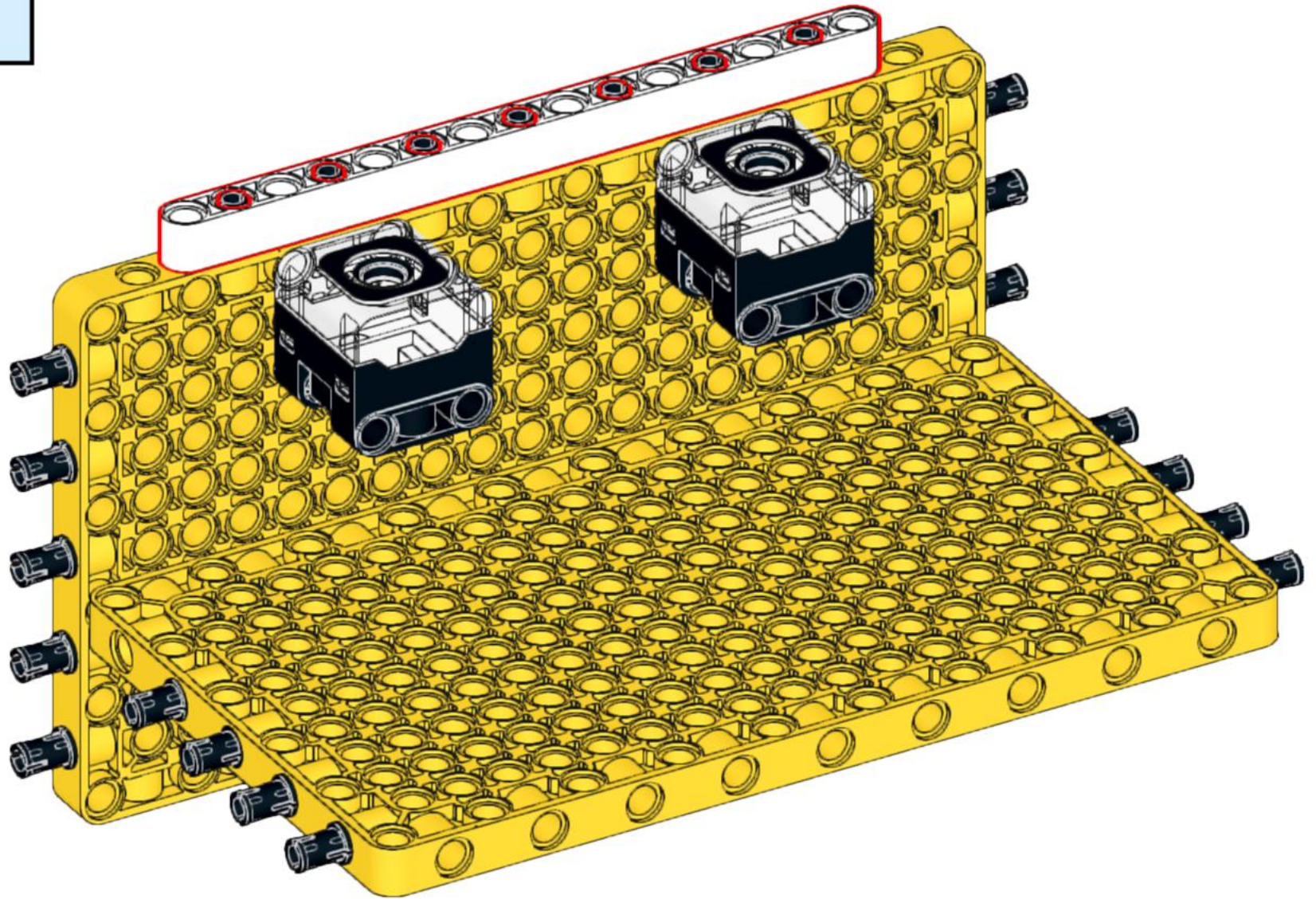
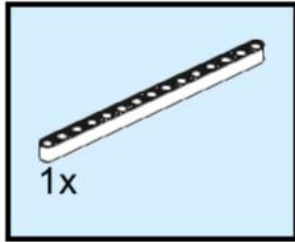
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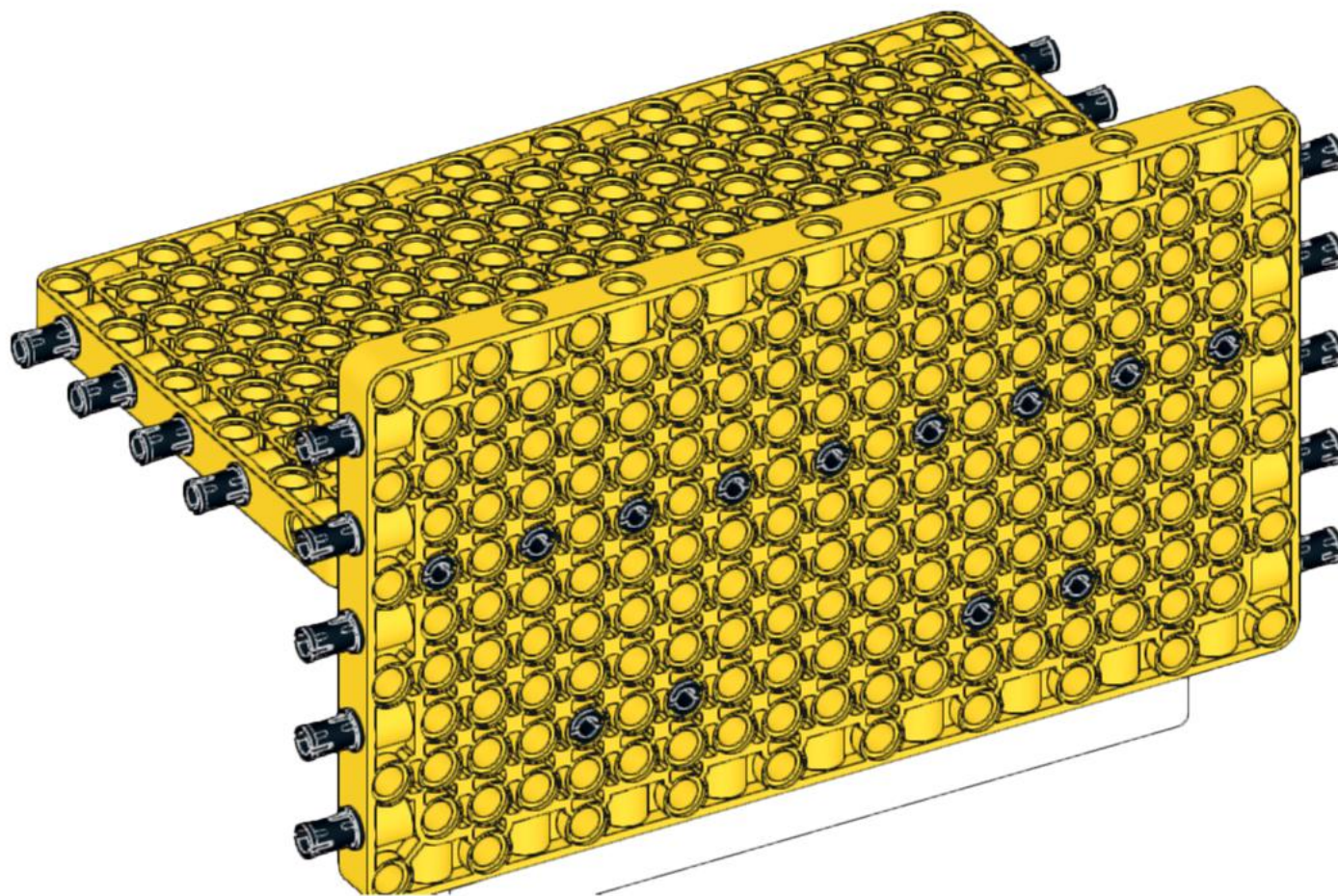
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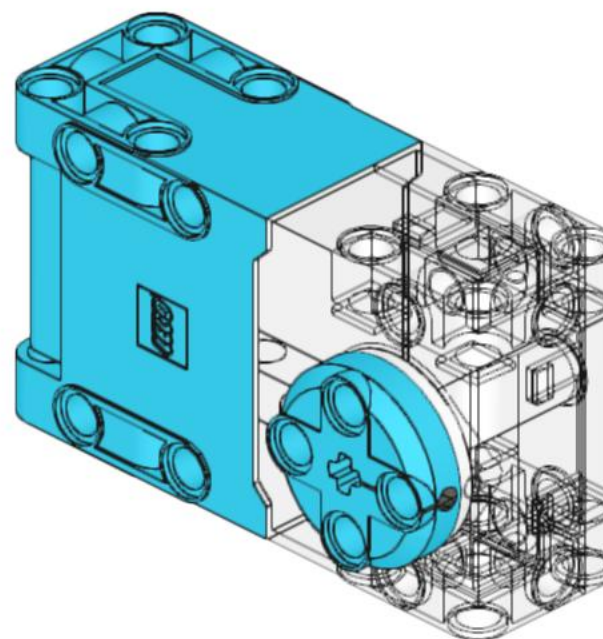
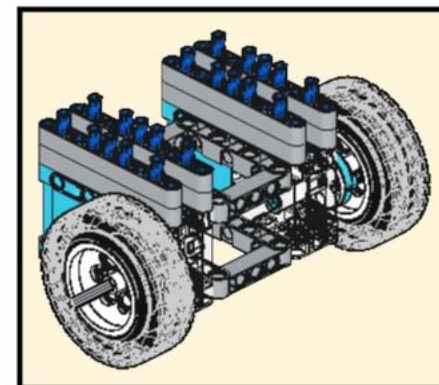
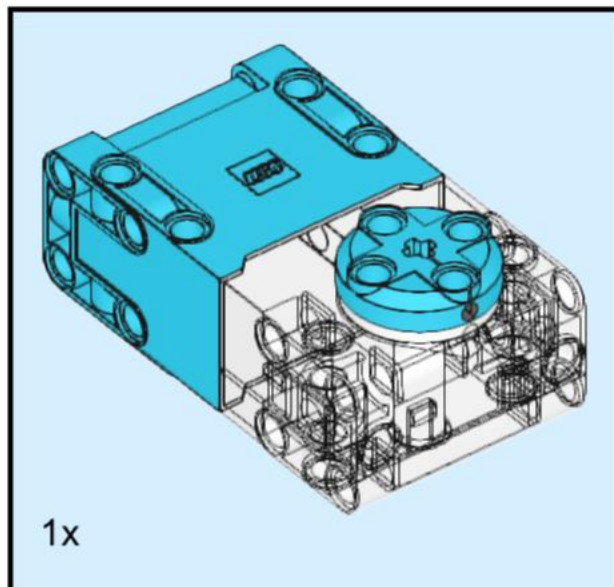
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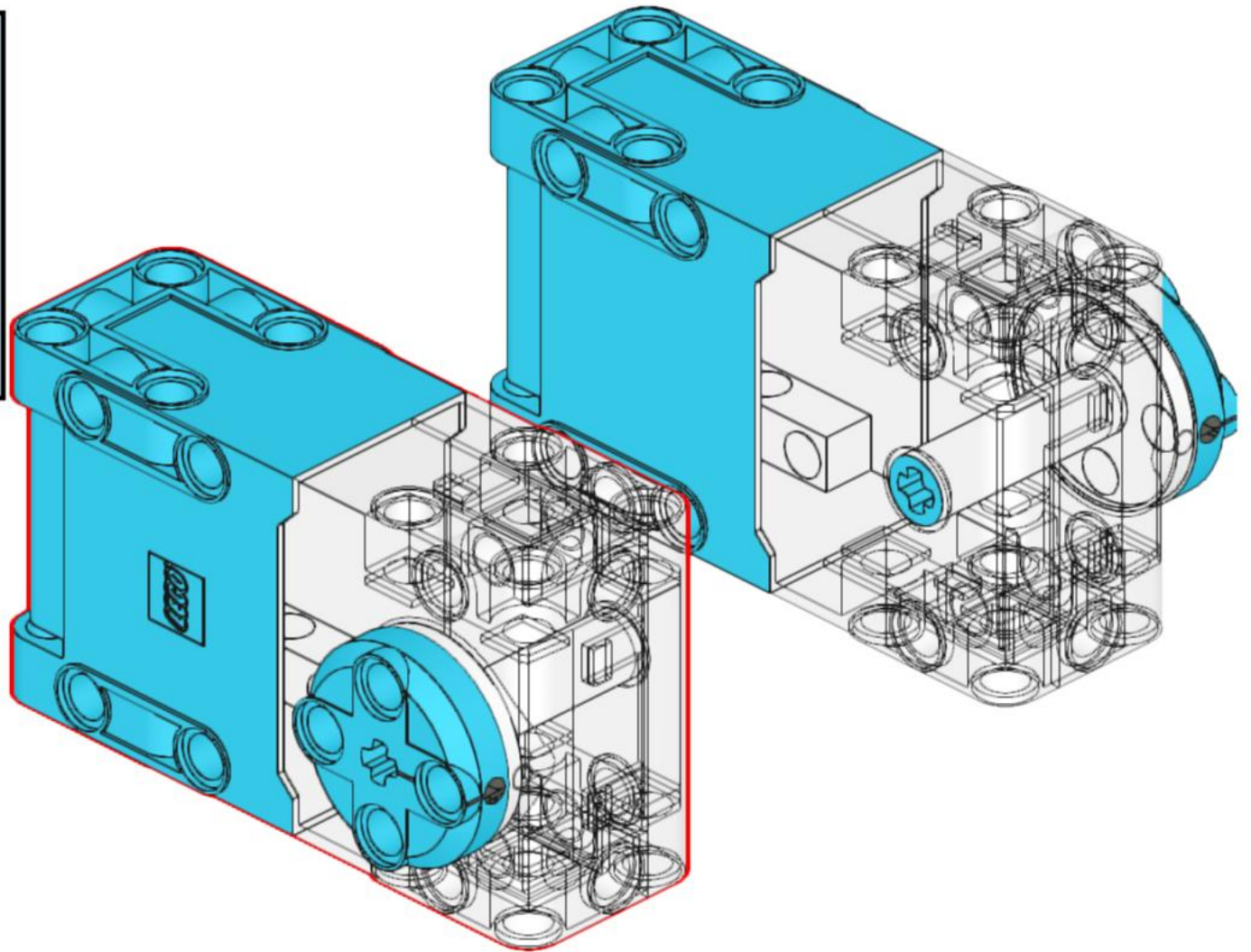
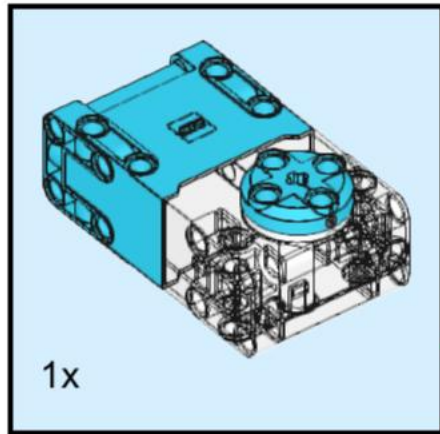
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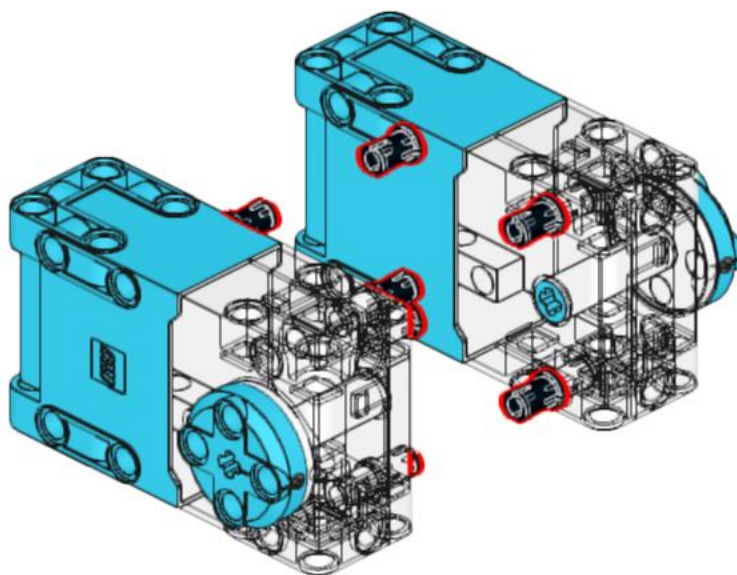
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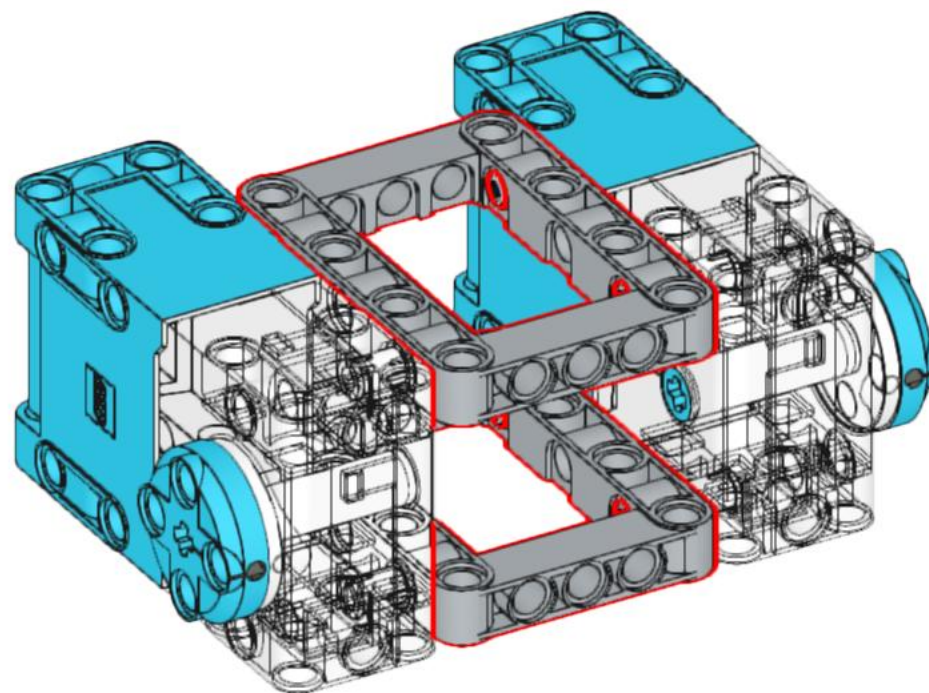
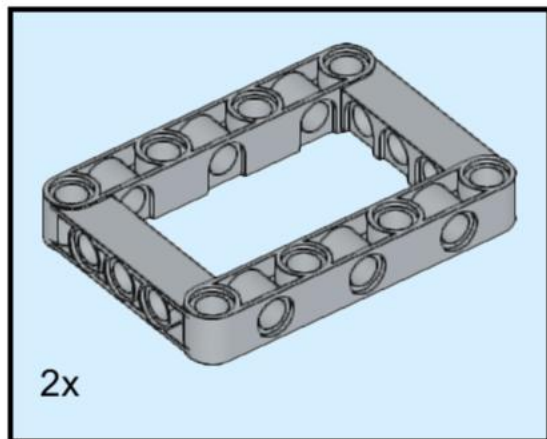
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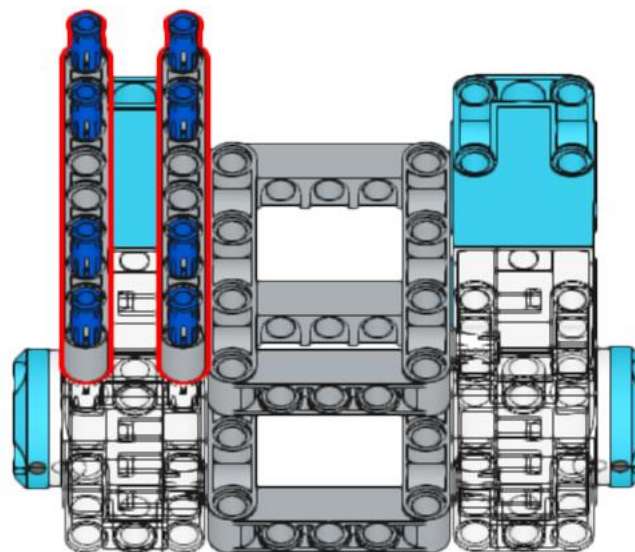
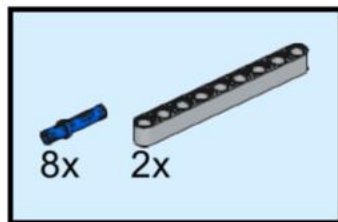
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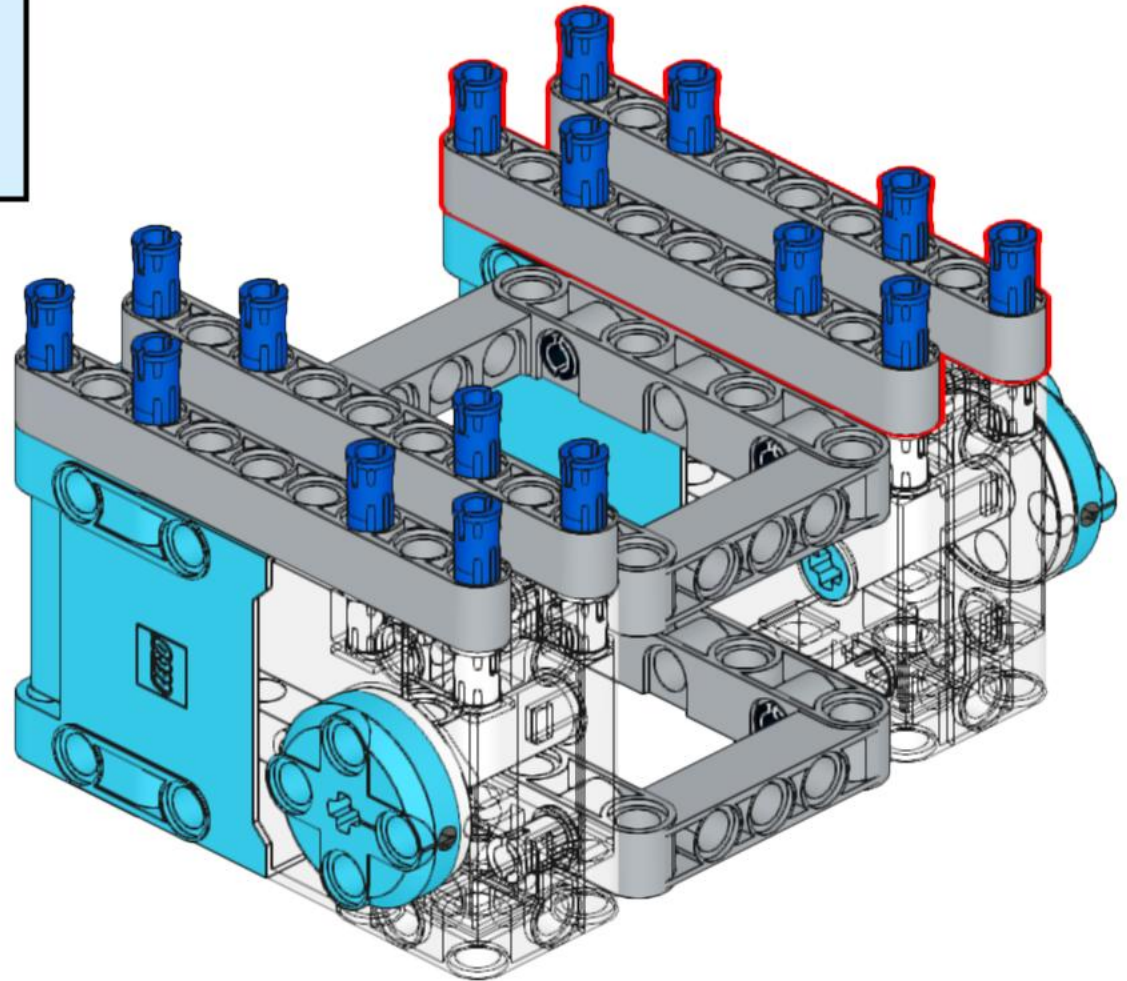
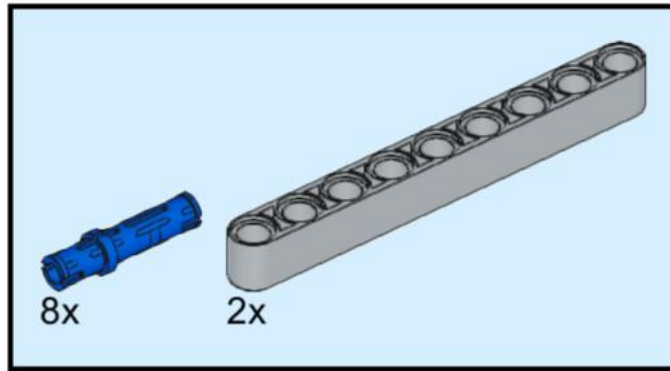
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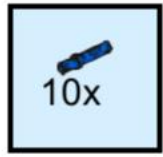
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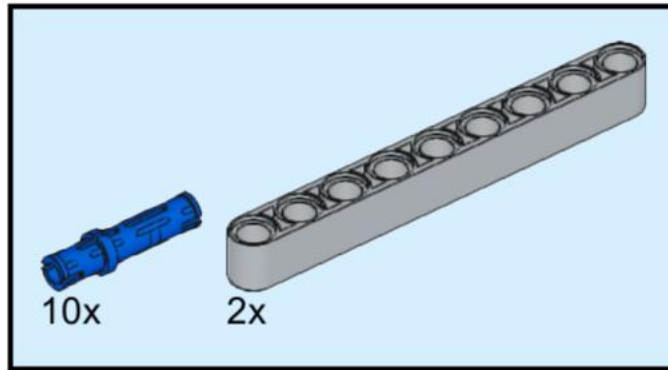
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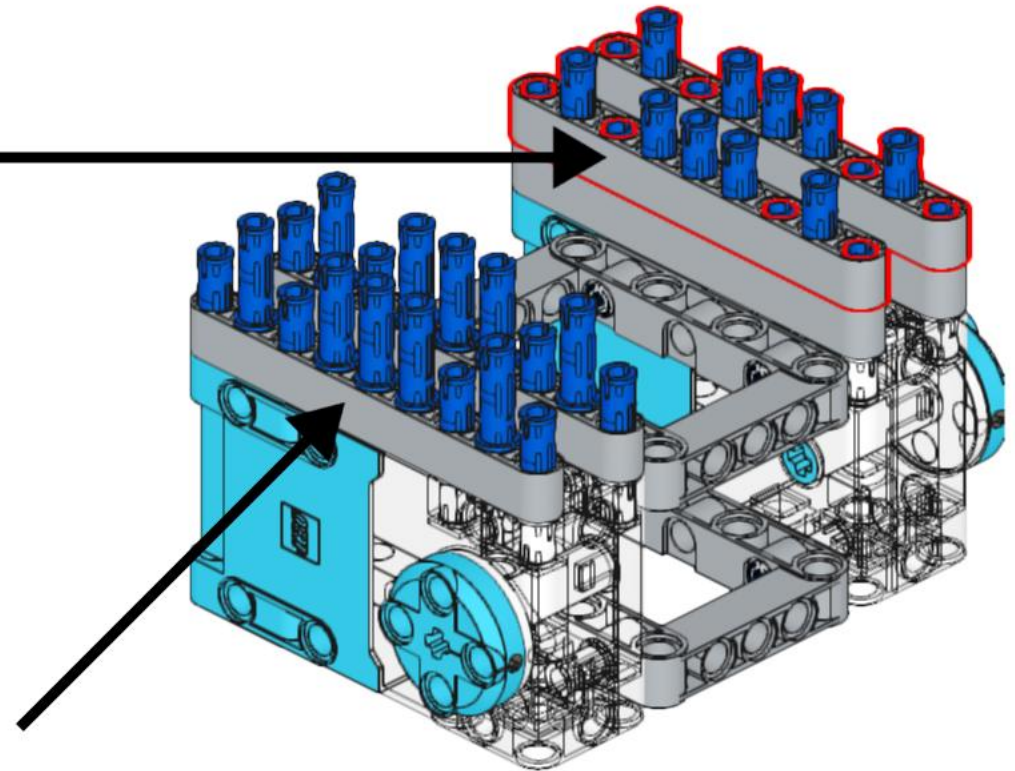
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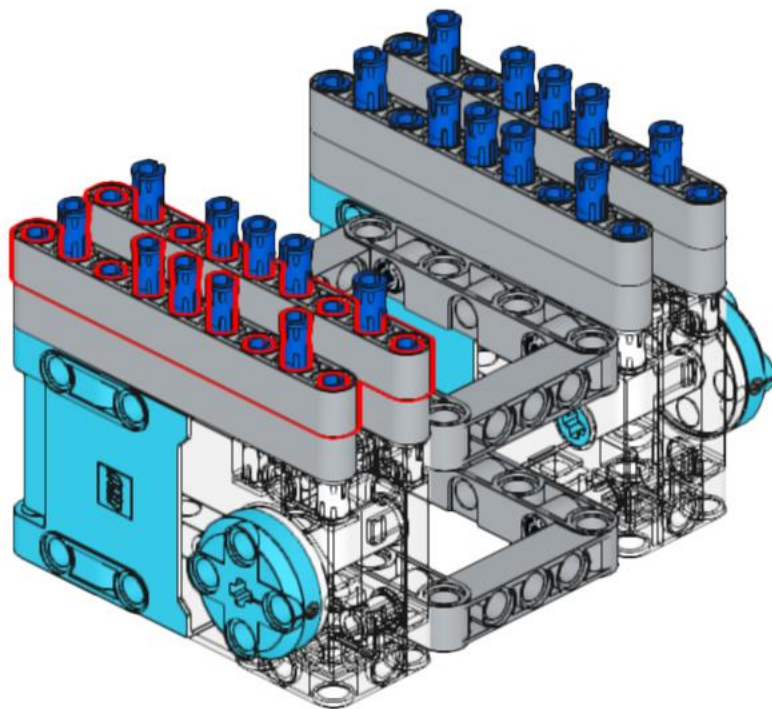
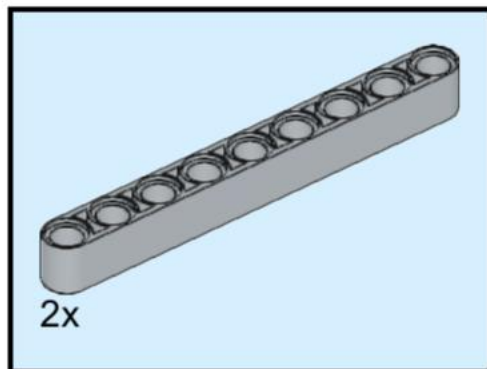
You might need to change which 3L pins are present depending on how your left and right attachments align up on the 11x19 plate that sits on top of the drive motors.

Another option is instead of recessing all the 3L Pins into the bottom gray beam, you can instead connect them to the top gray beam so they will penetrate the 11x19 plate and connect to you attachment motors.

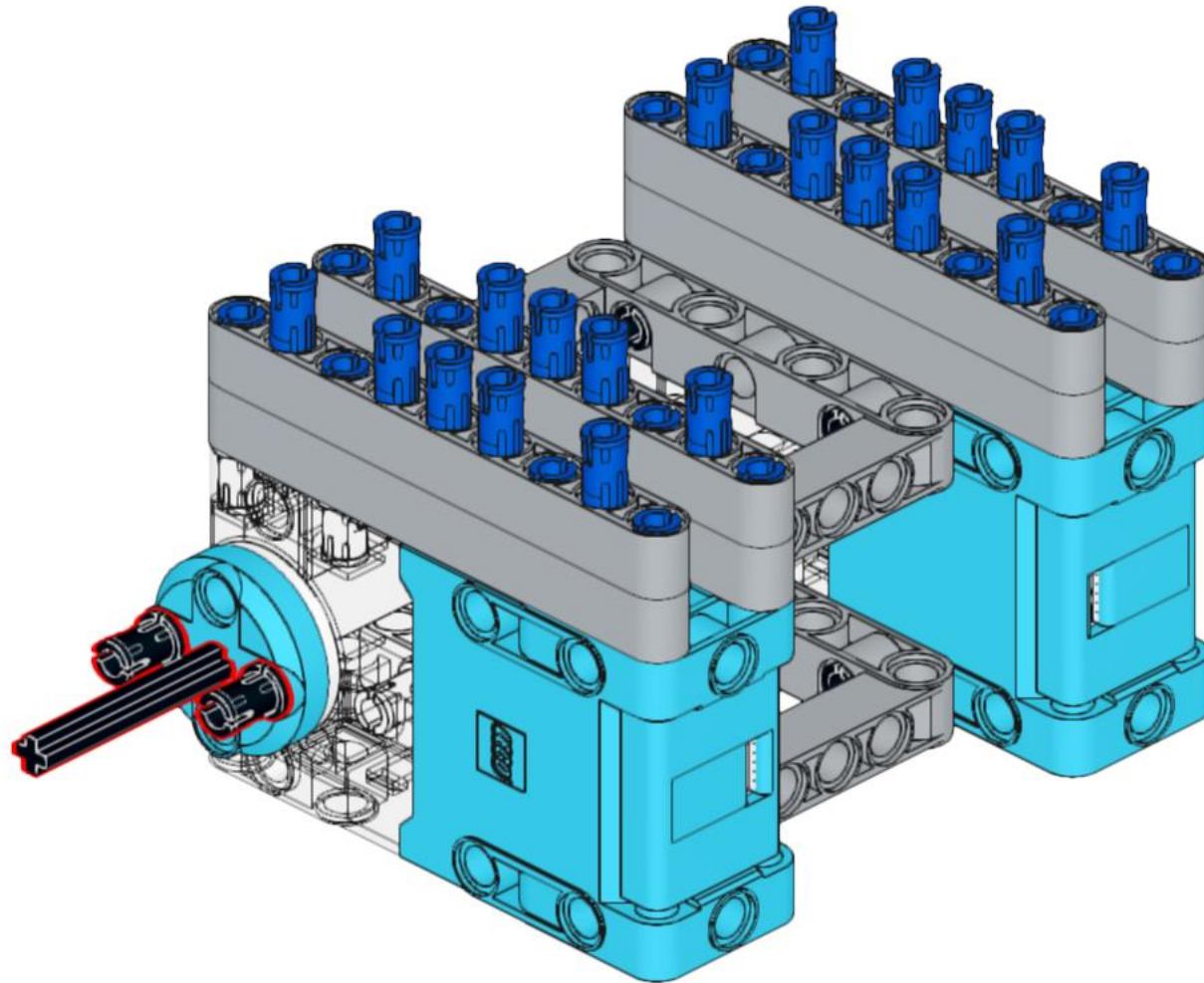
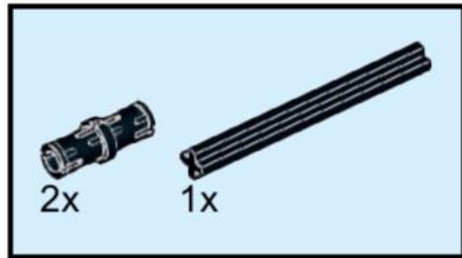
It is a tradeoff in strength vs. flexibility.



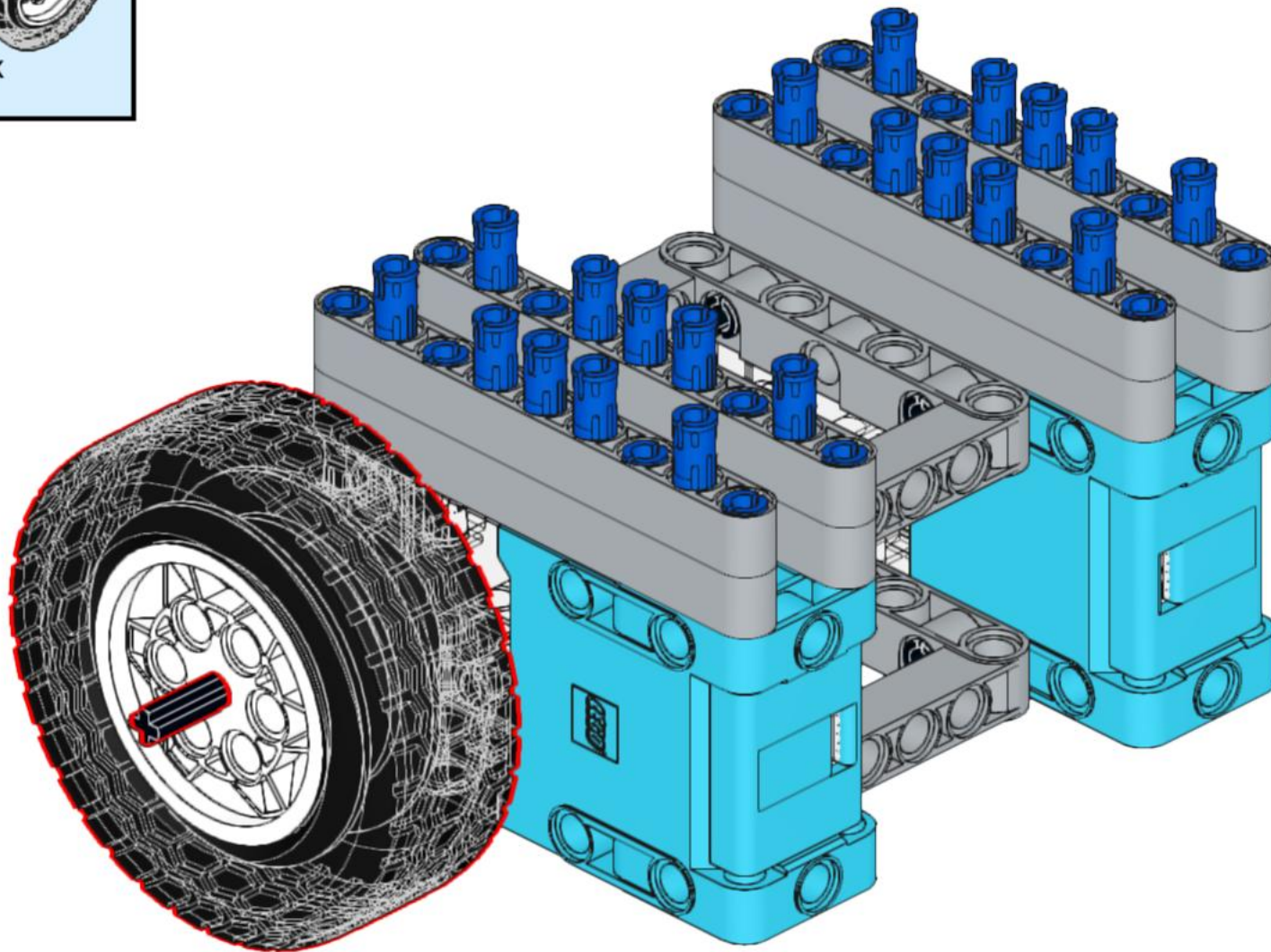
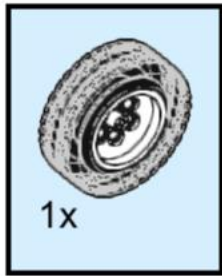
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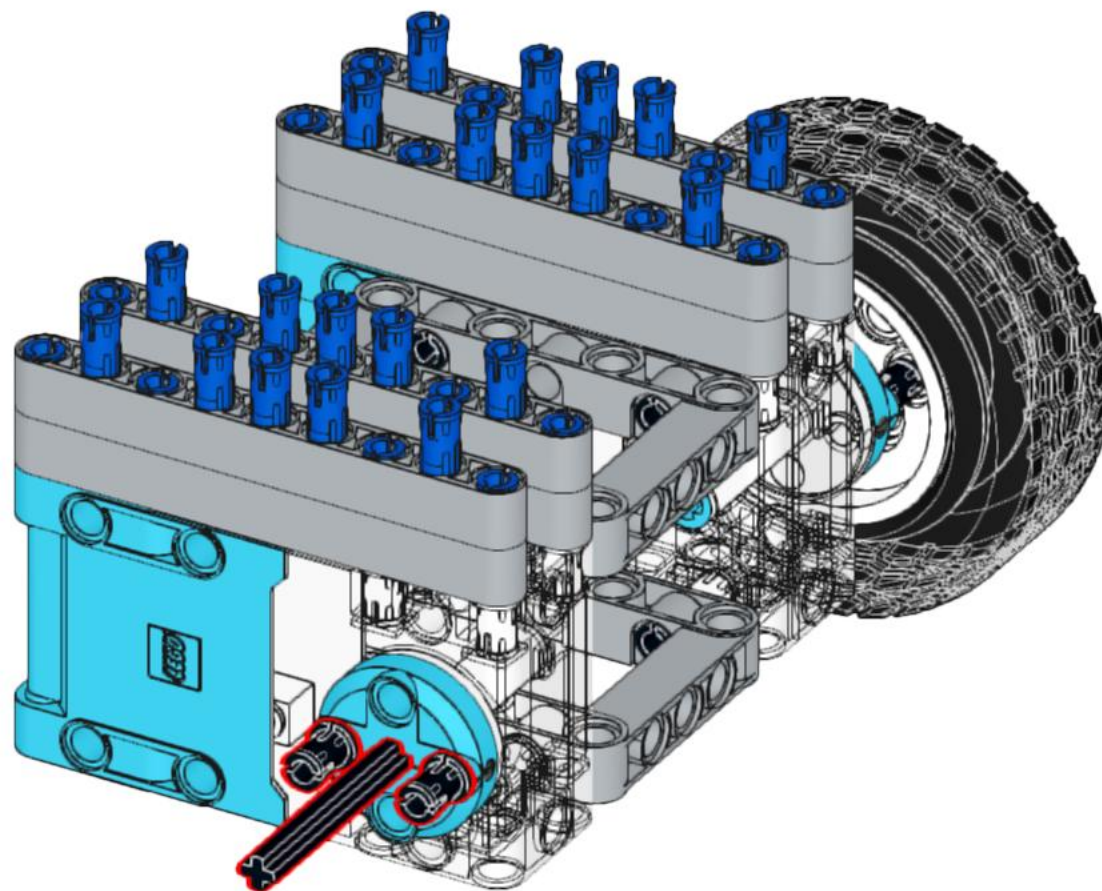
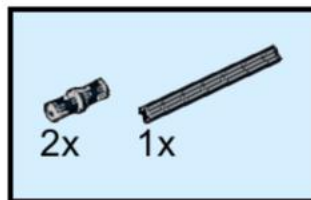
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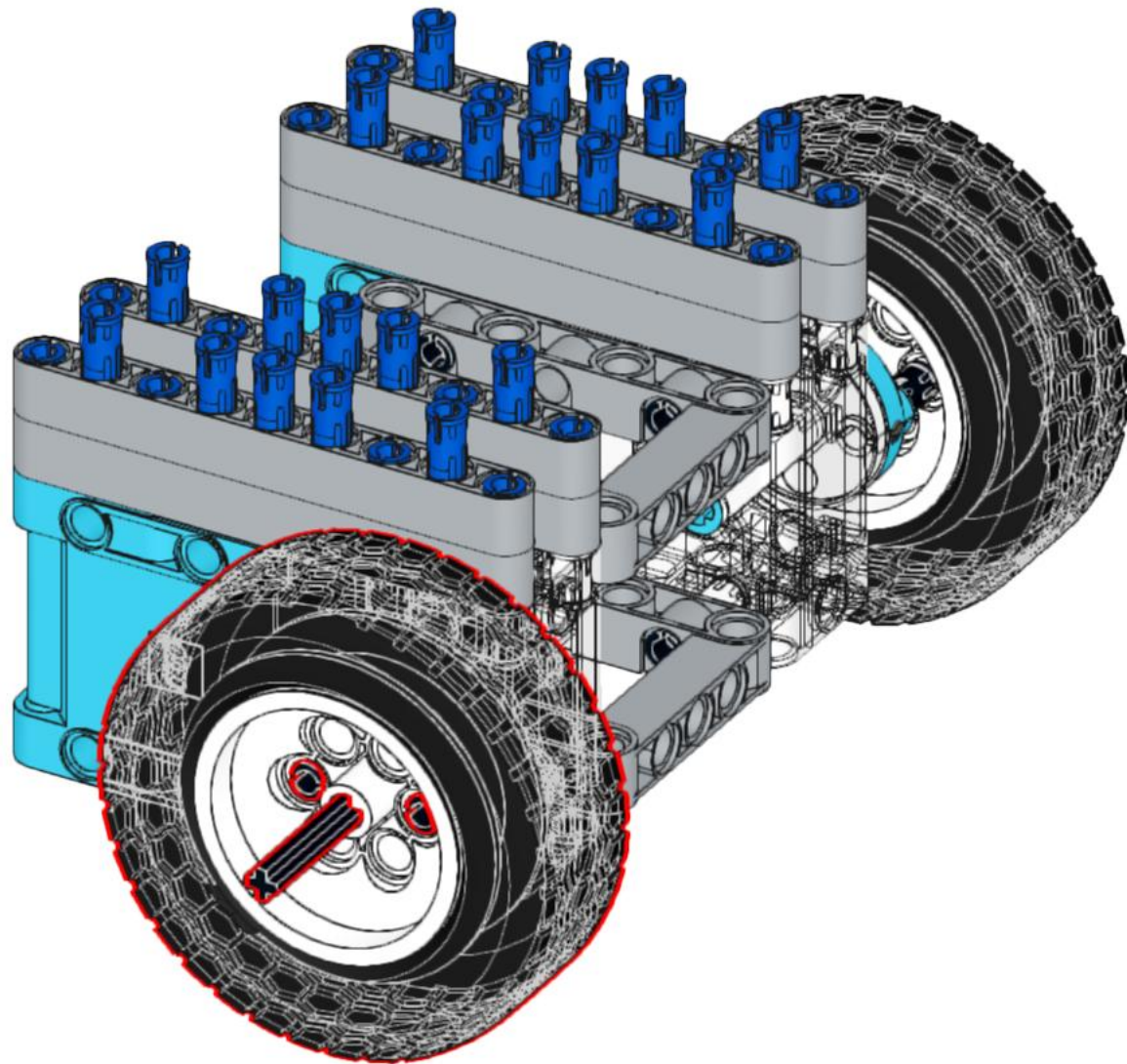
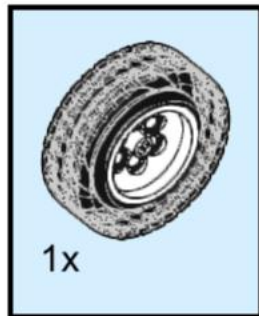
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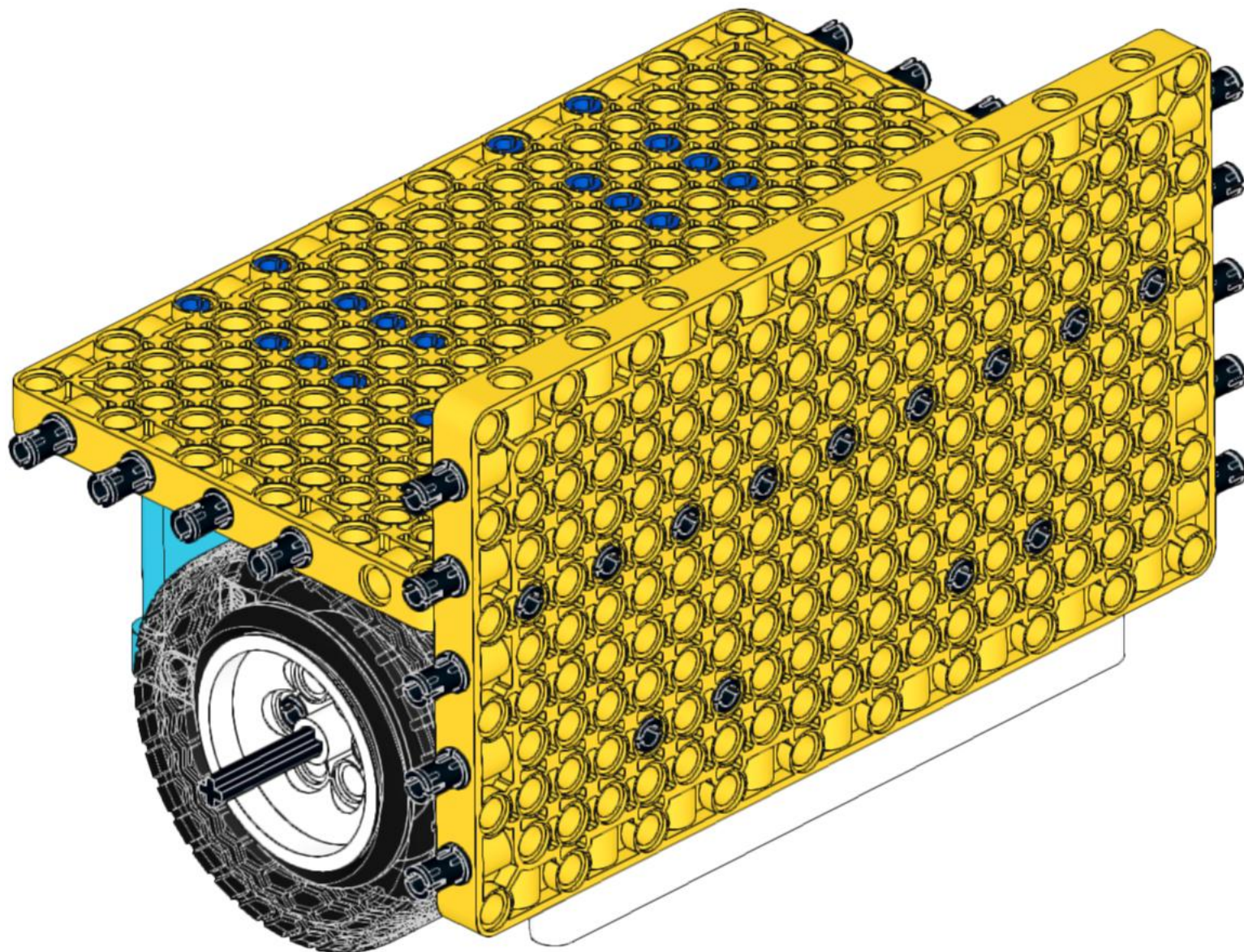
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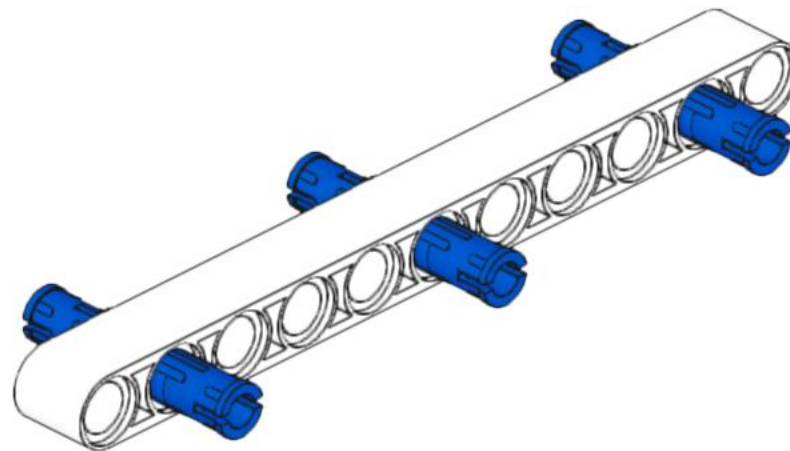
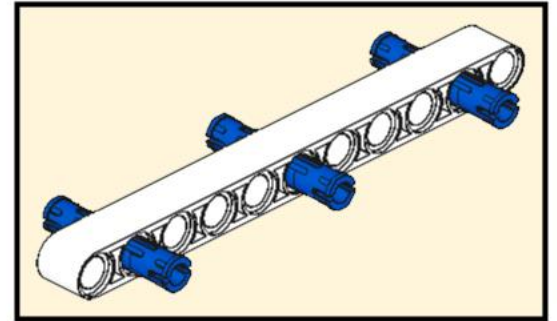
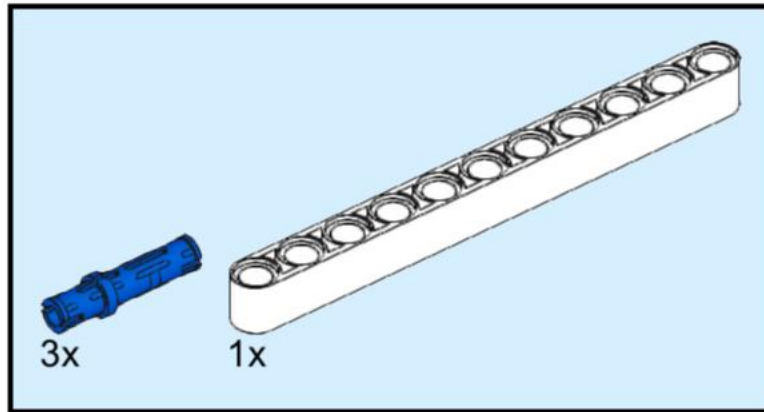


Rear Assembly

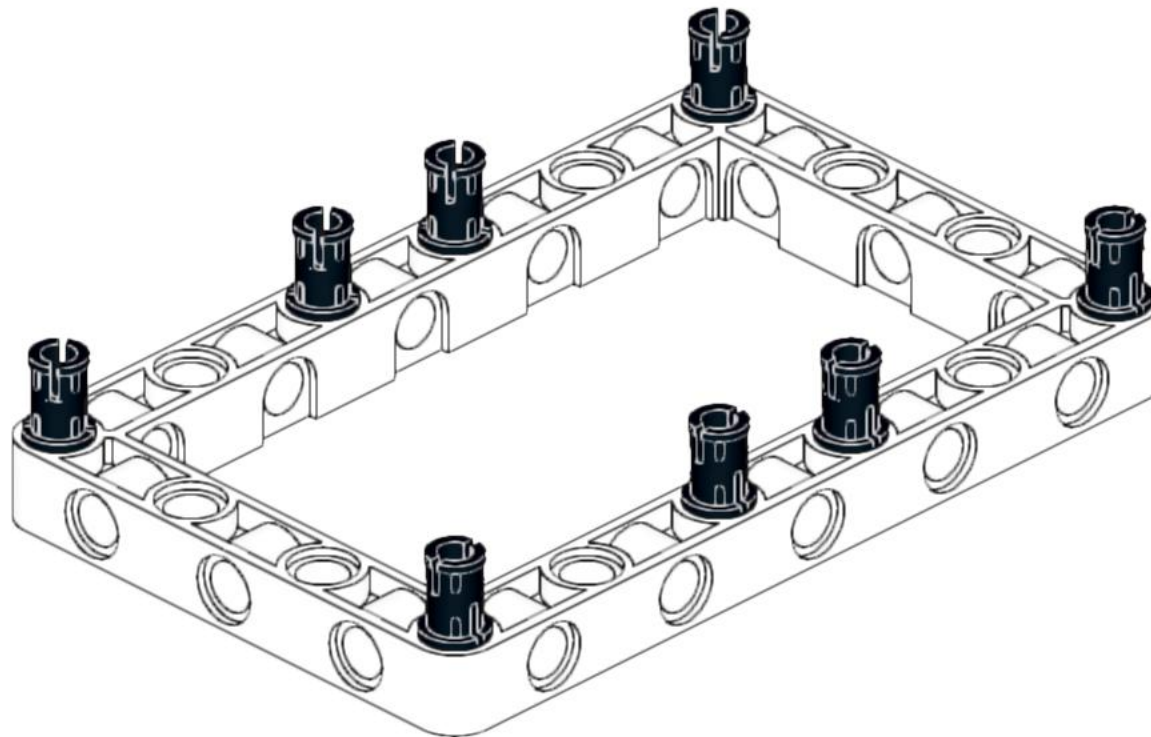
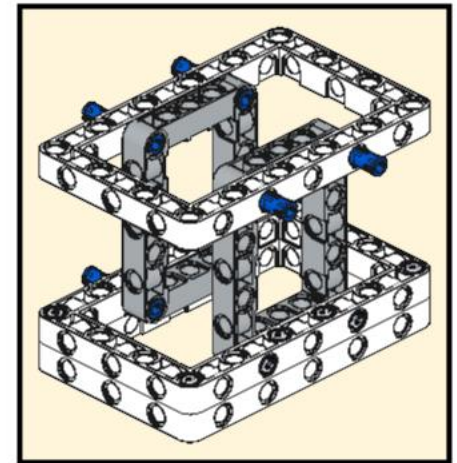
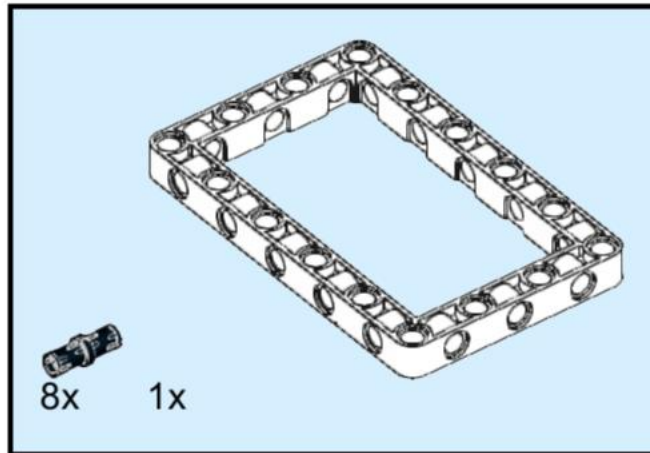
This is the portion of the robot that contains the hub.

While not shown in these plans, we sometimes installed two 2×6×2 Brick, Counter Weights (73090) in the middle 5x7 frame to offset the weight of the front of the robot.

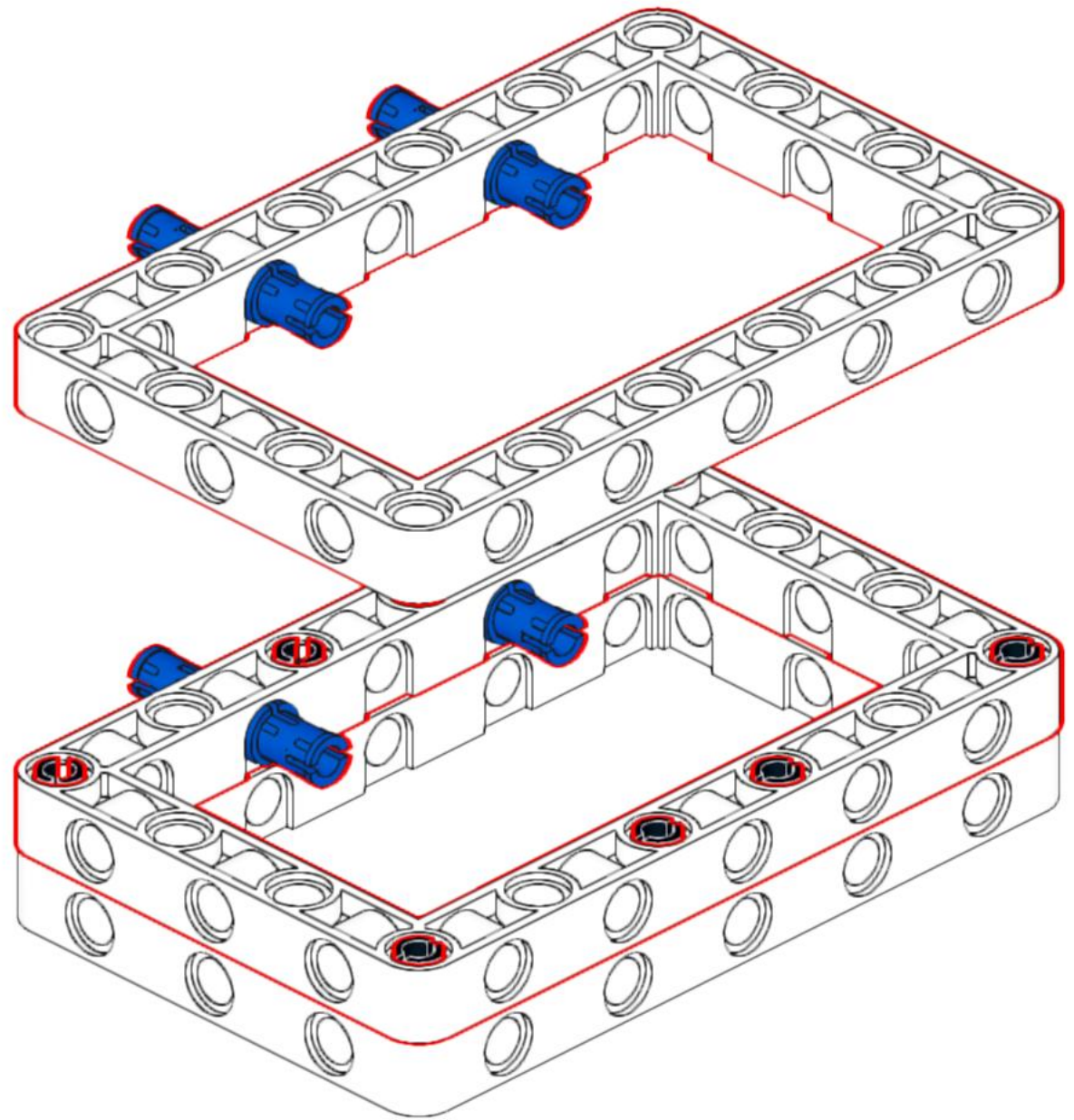
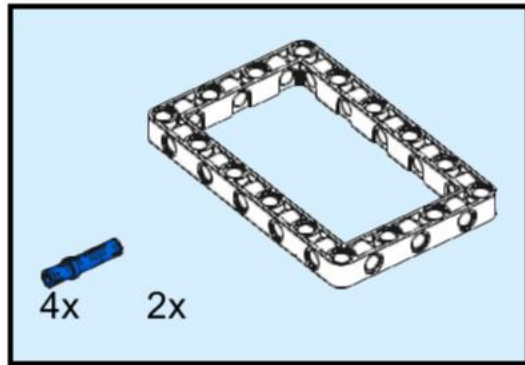
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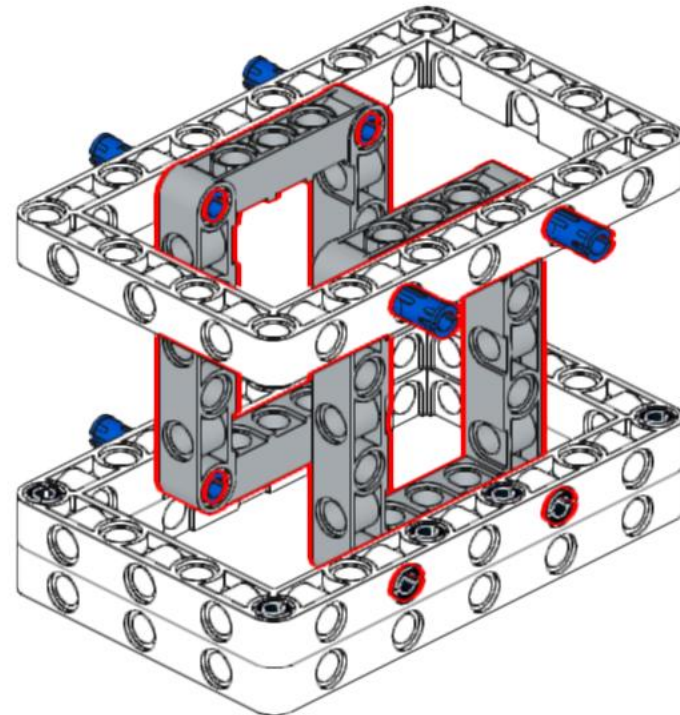
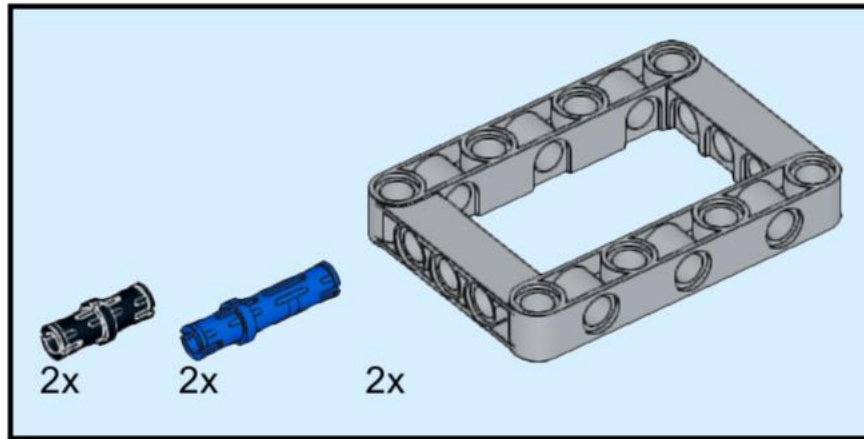
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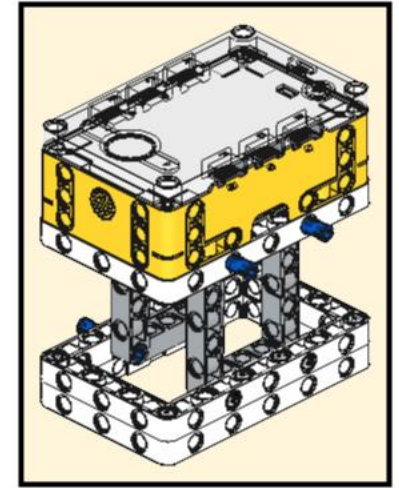
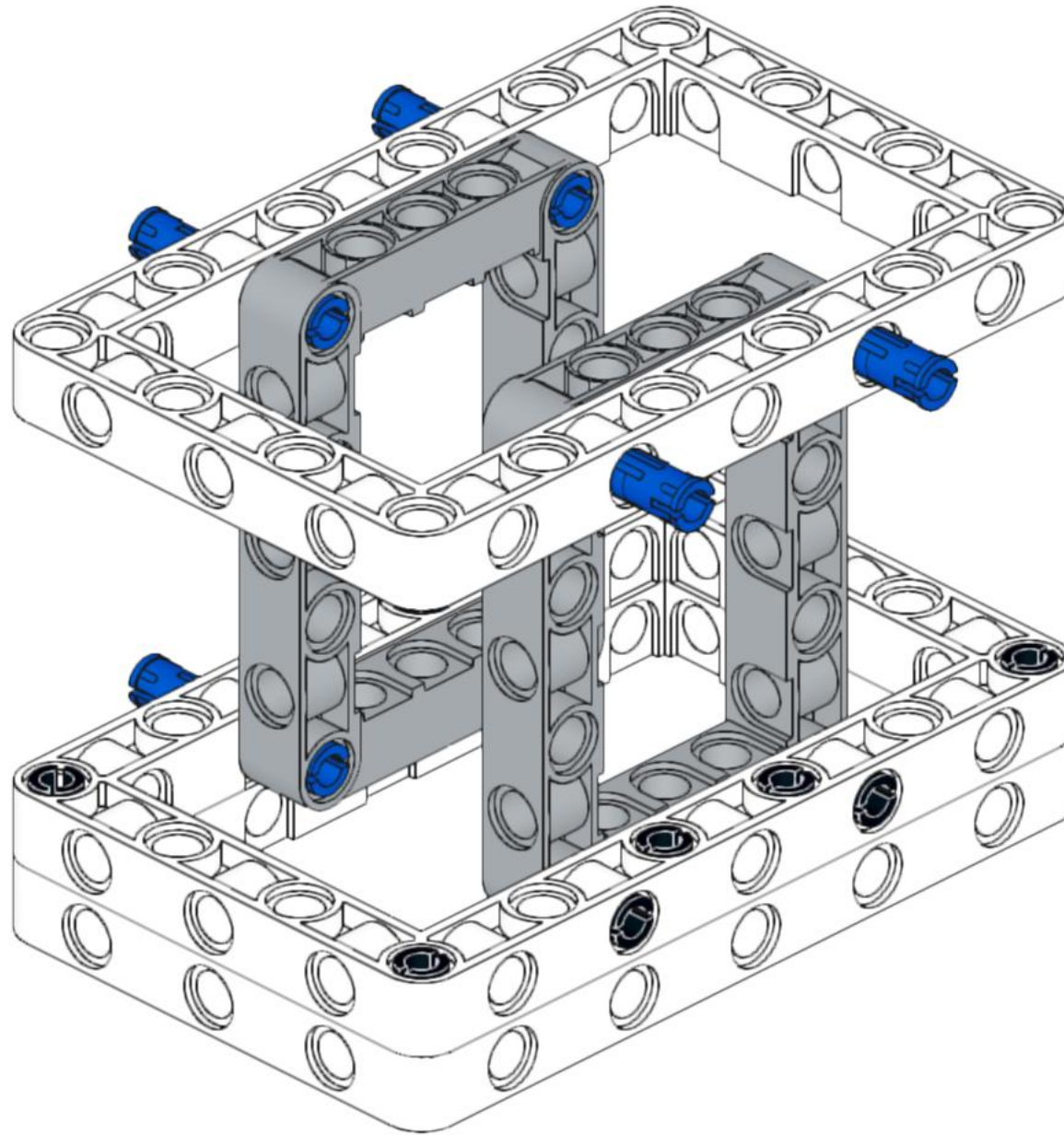
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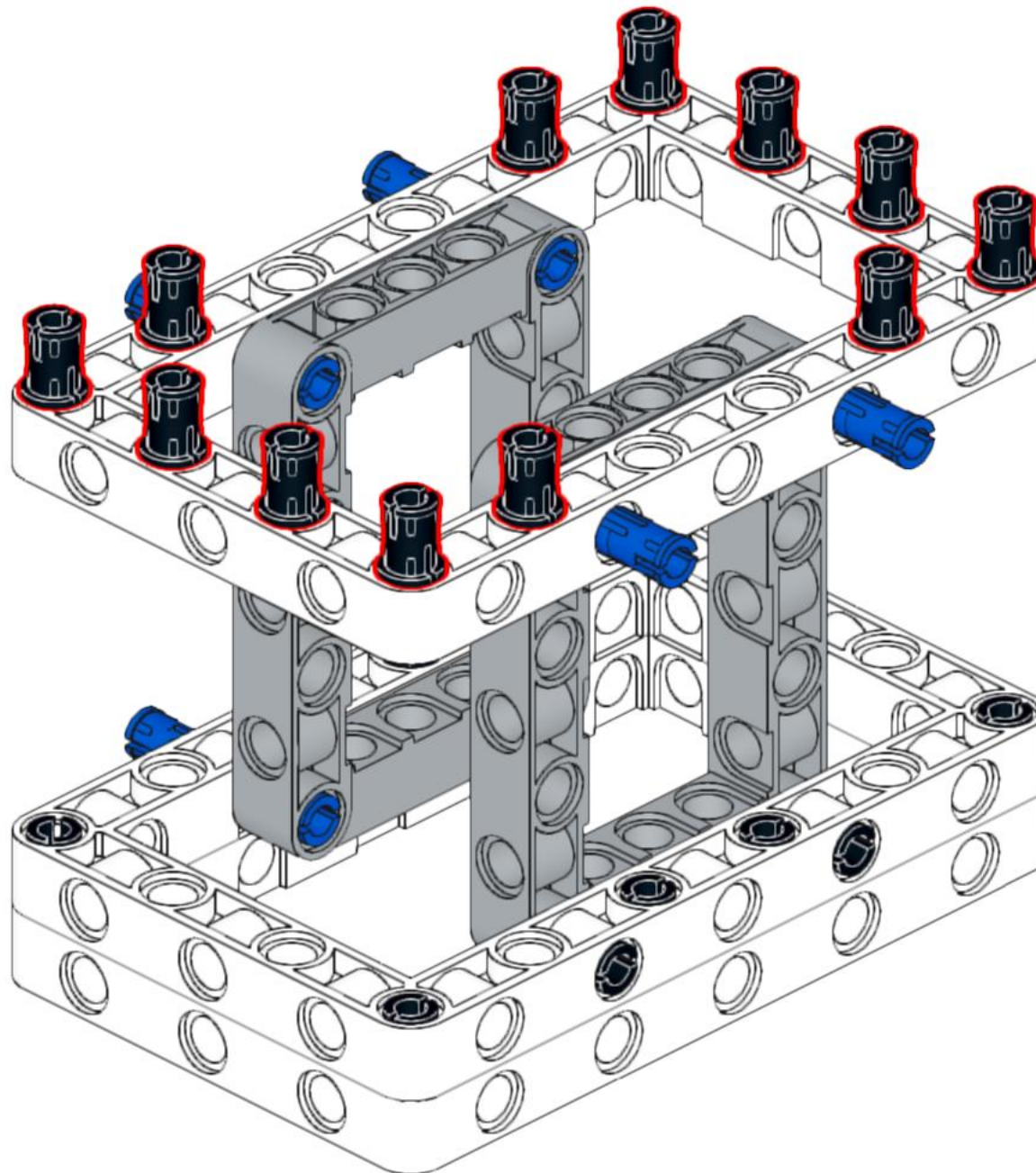
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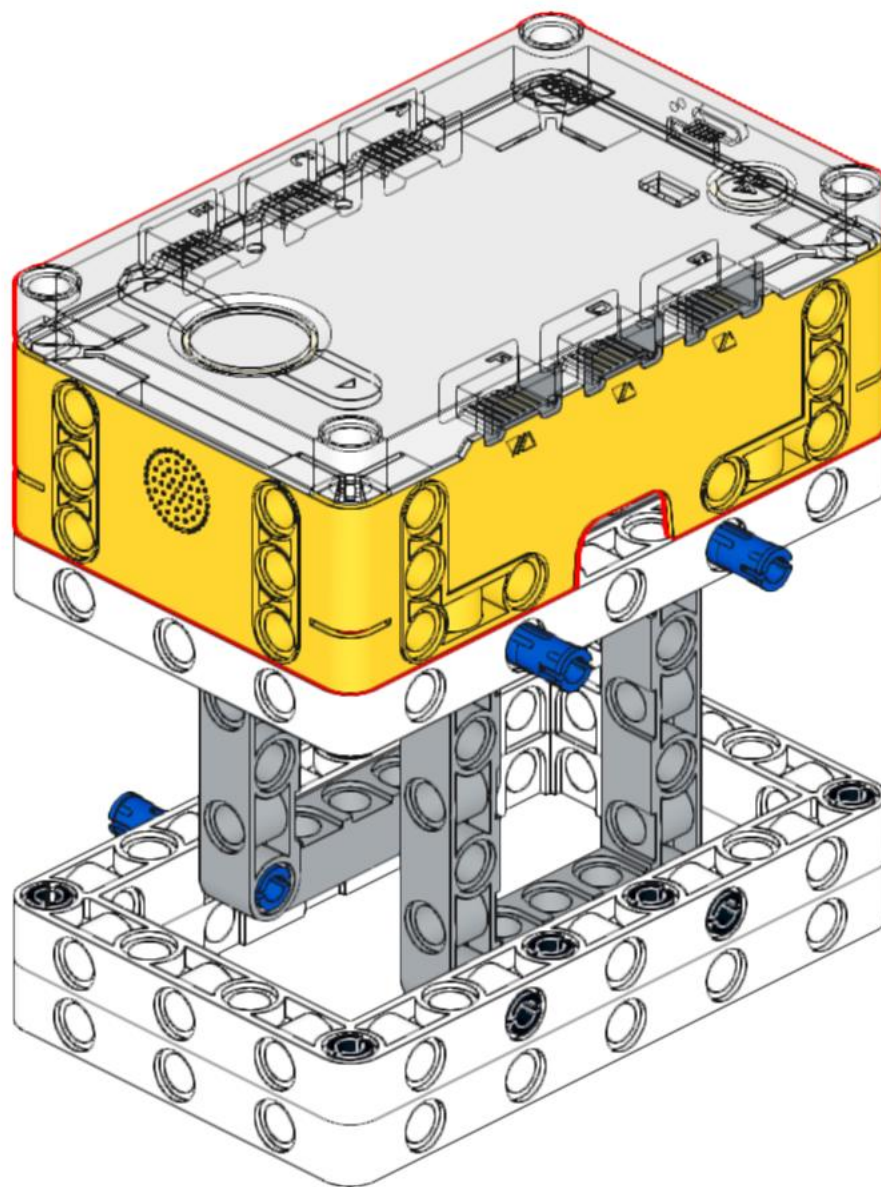
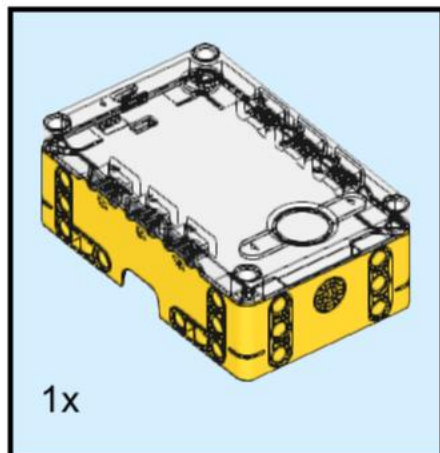
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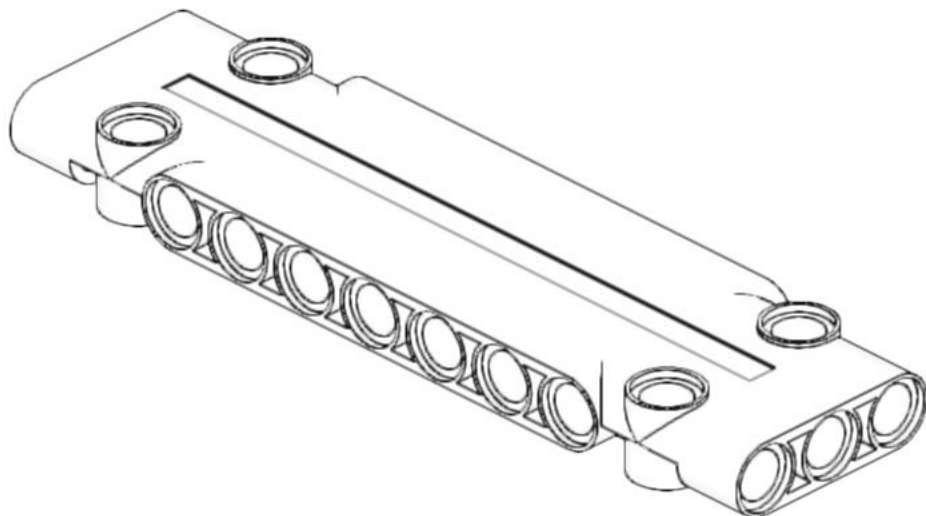
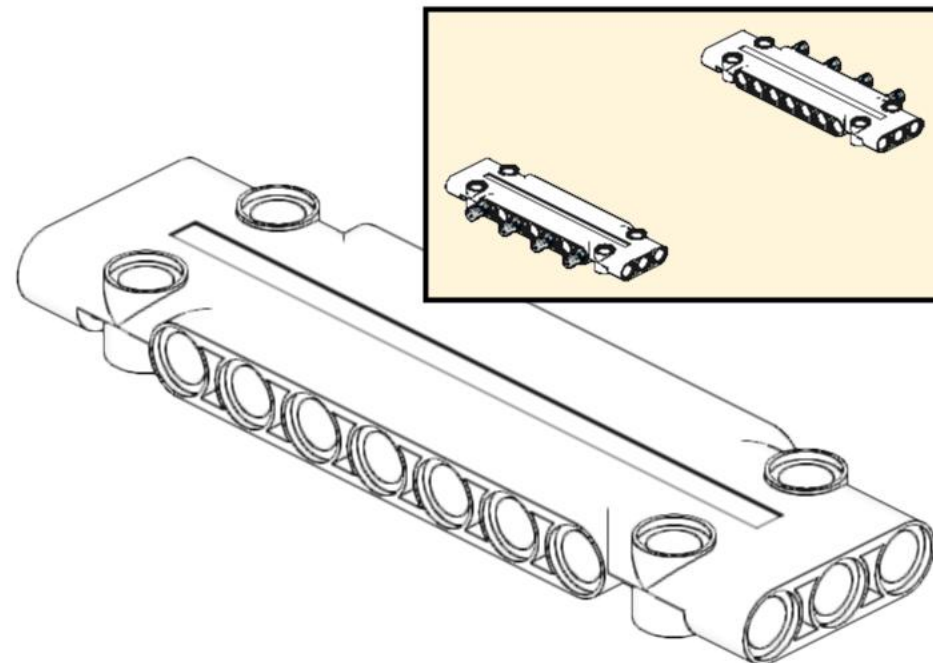
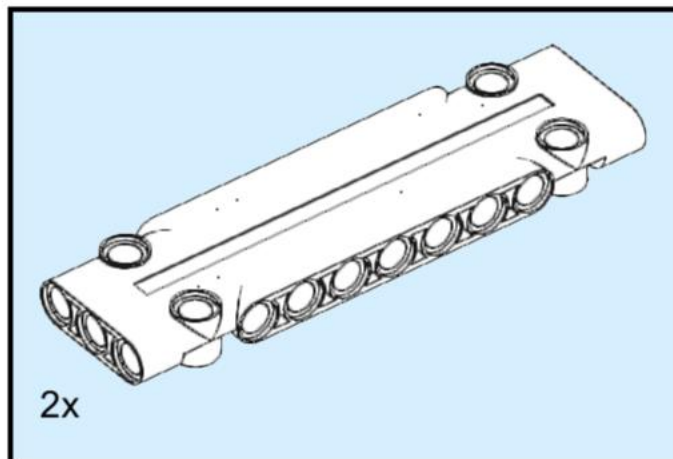
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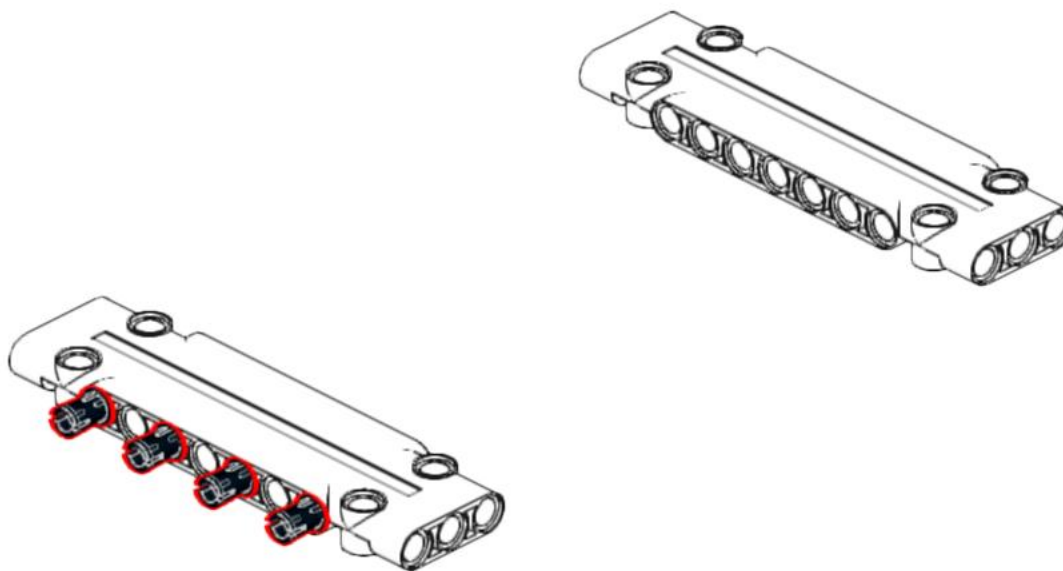
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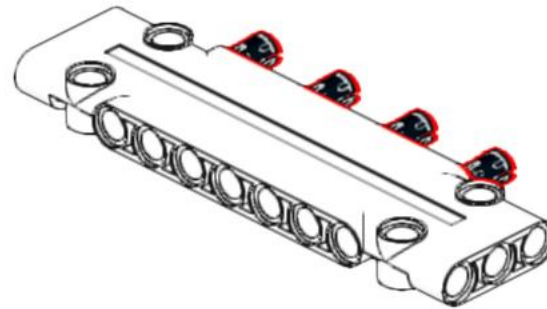
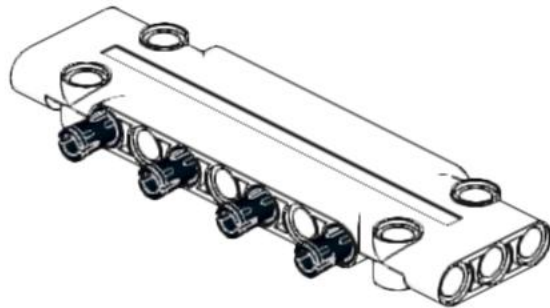
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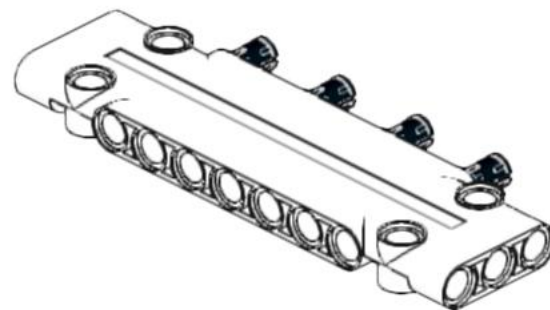
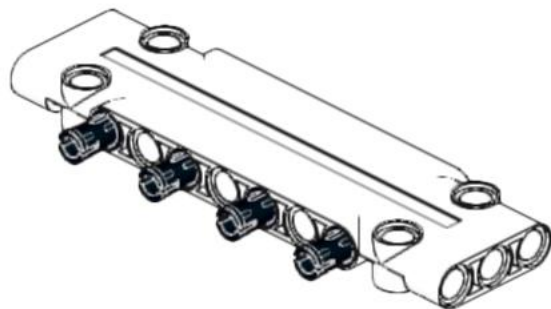
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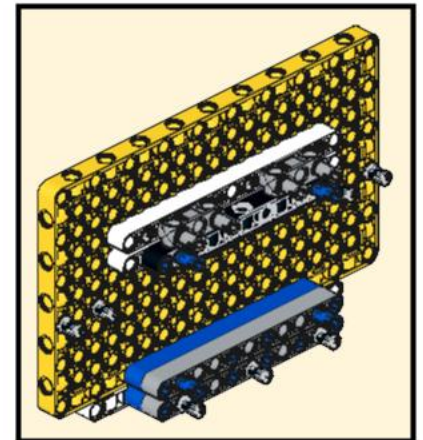
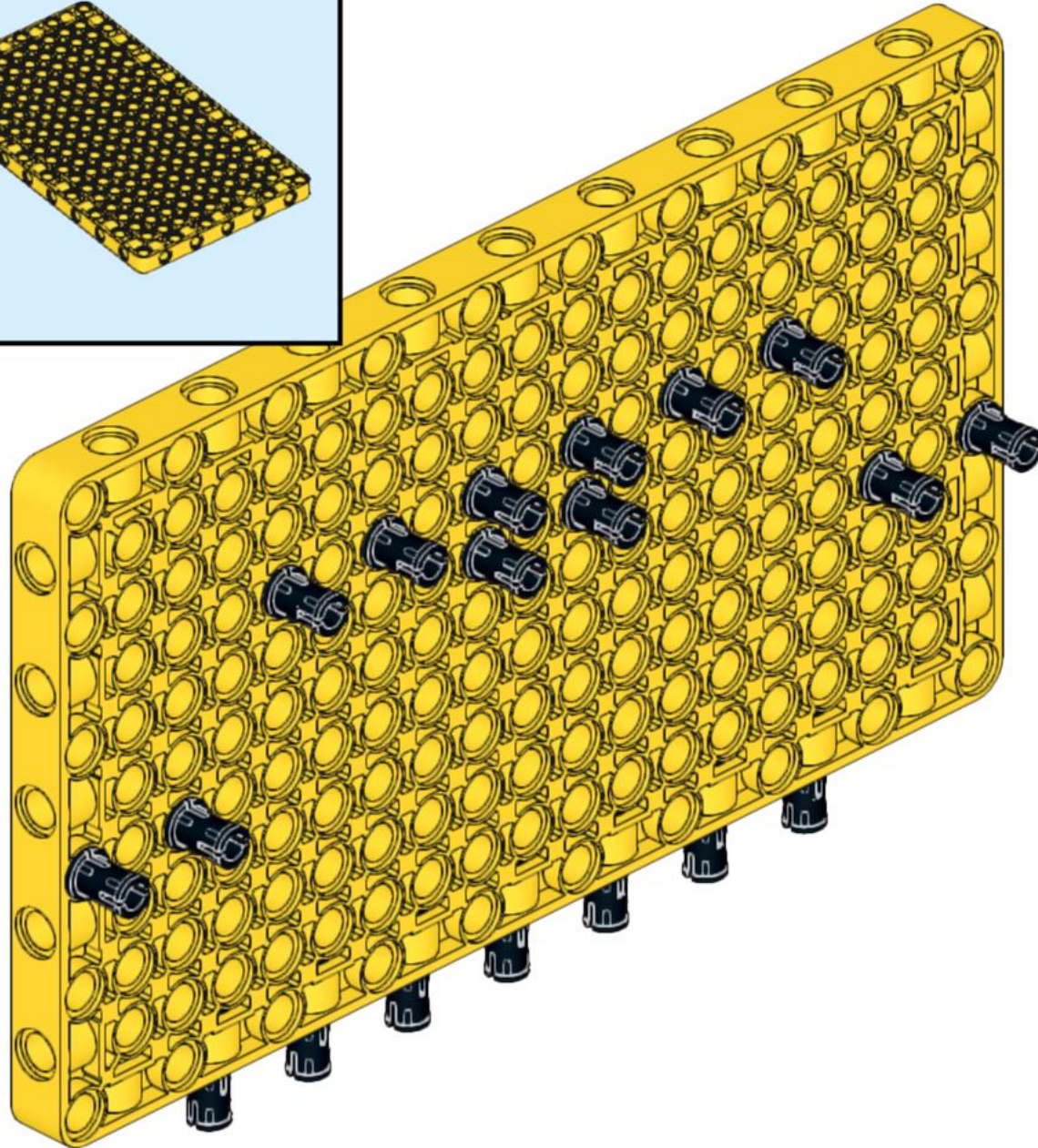
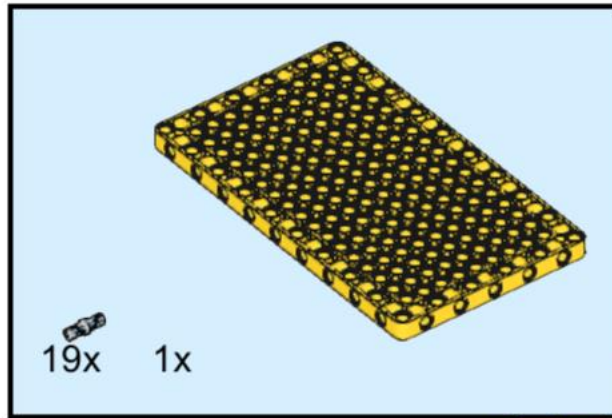
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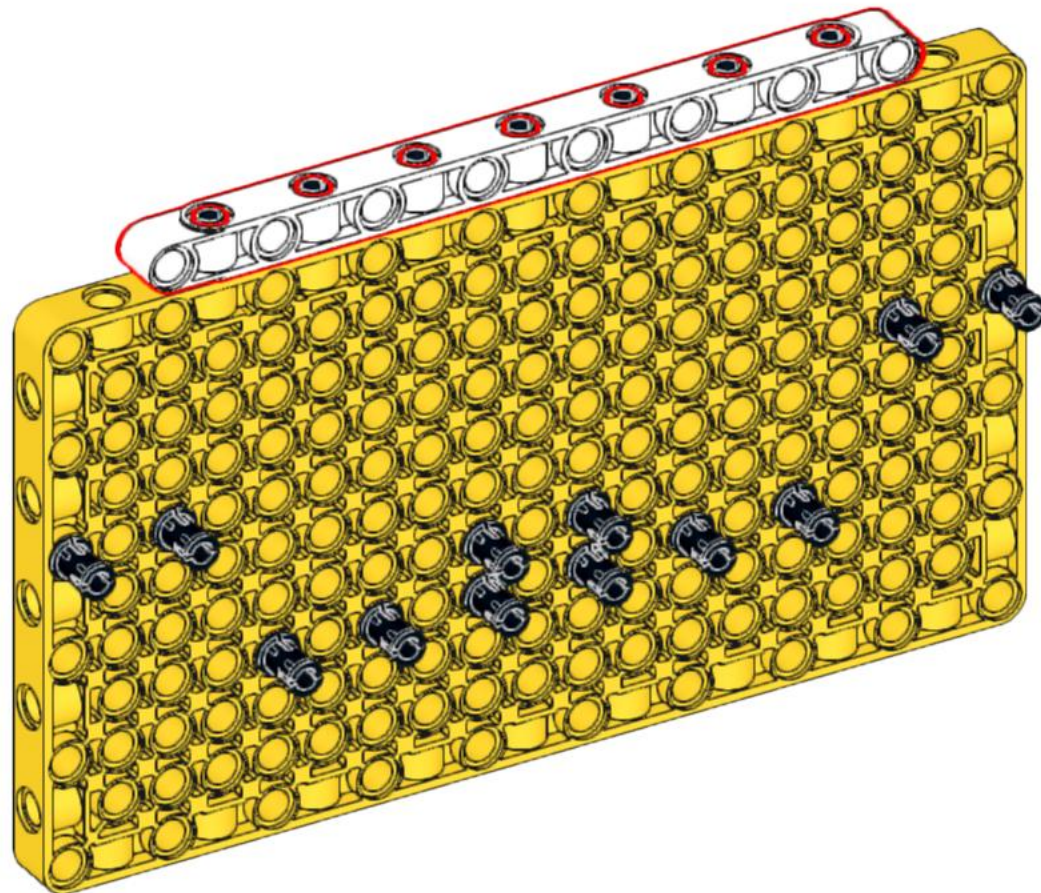
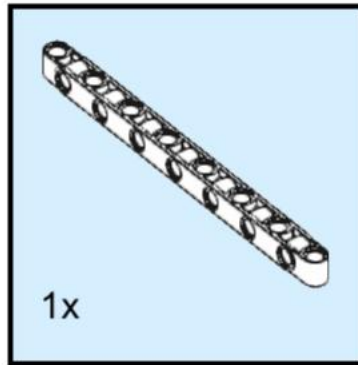
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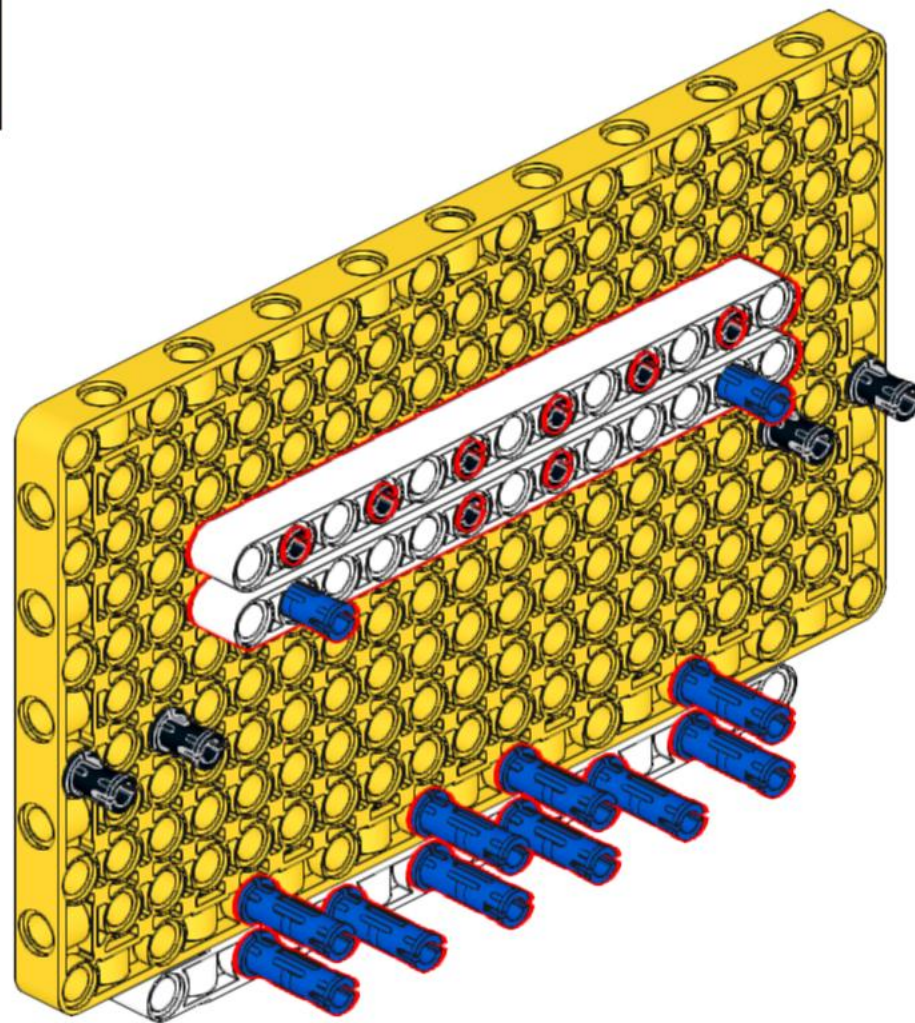
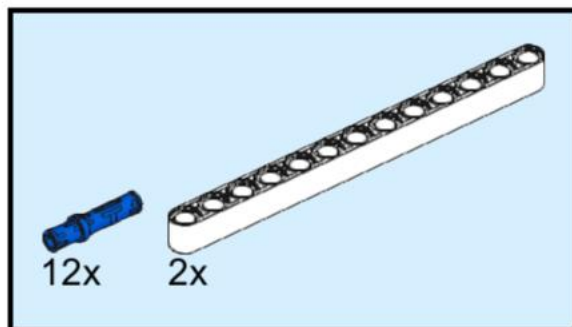
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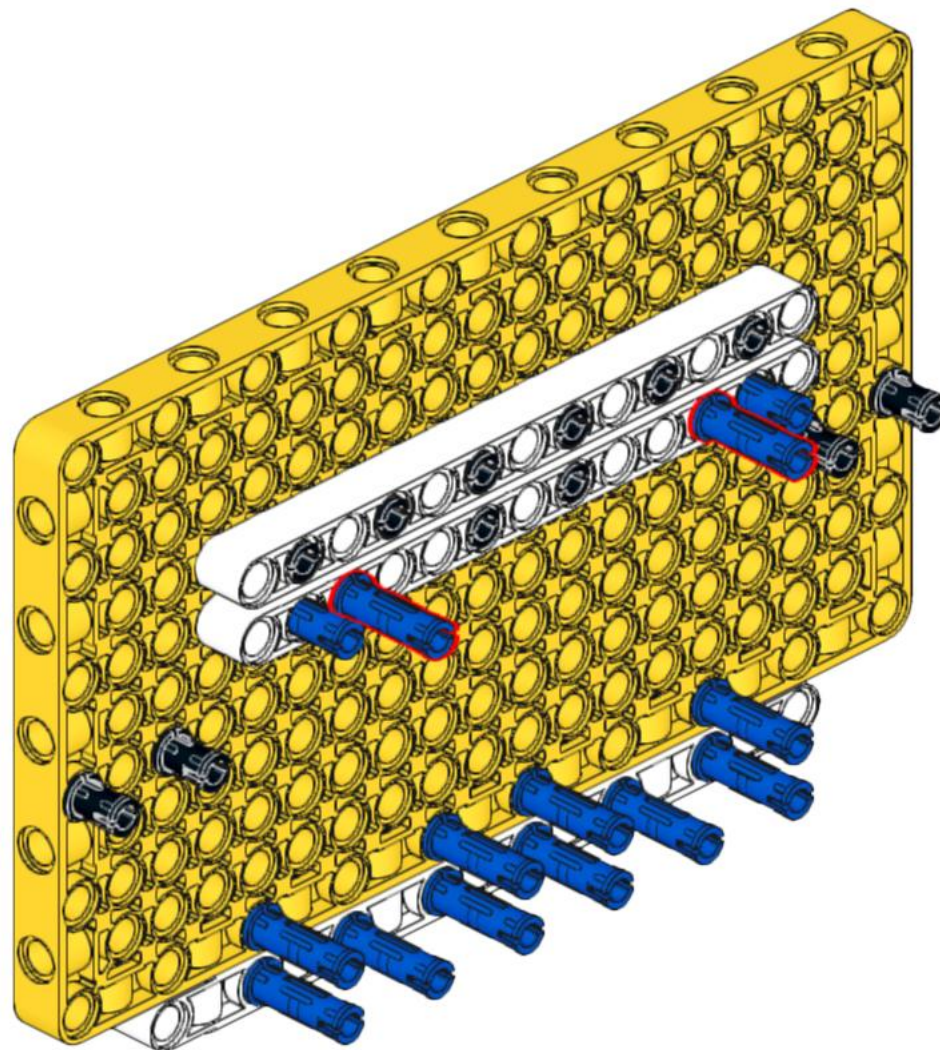
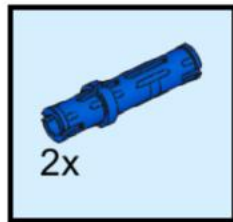
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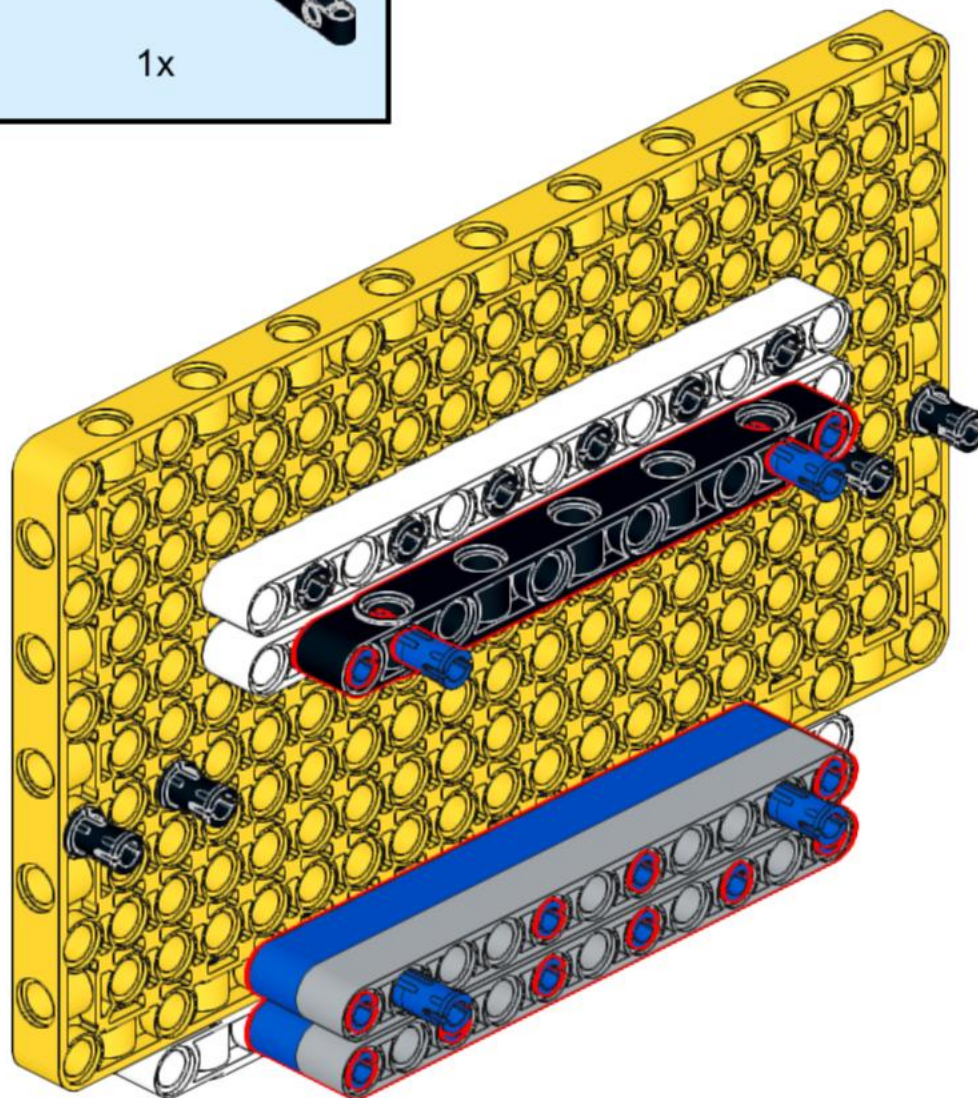
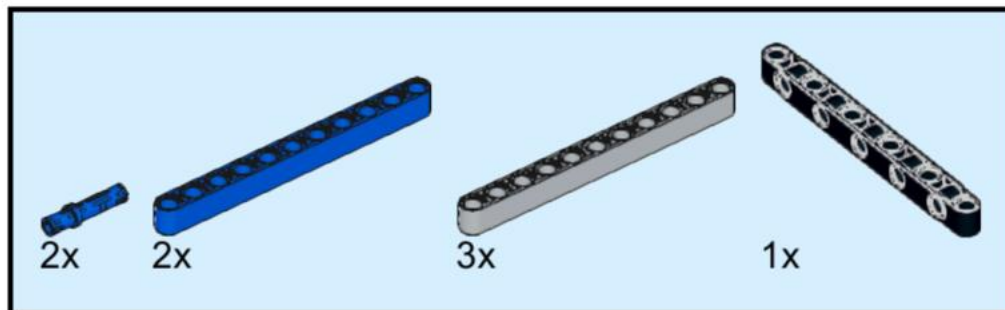
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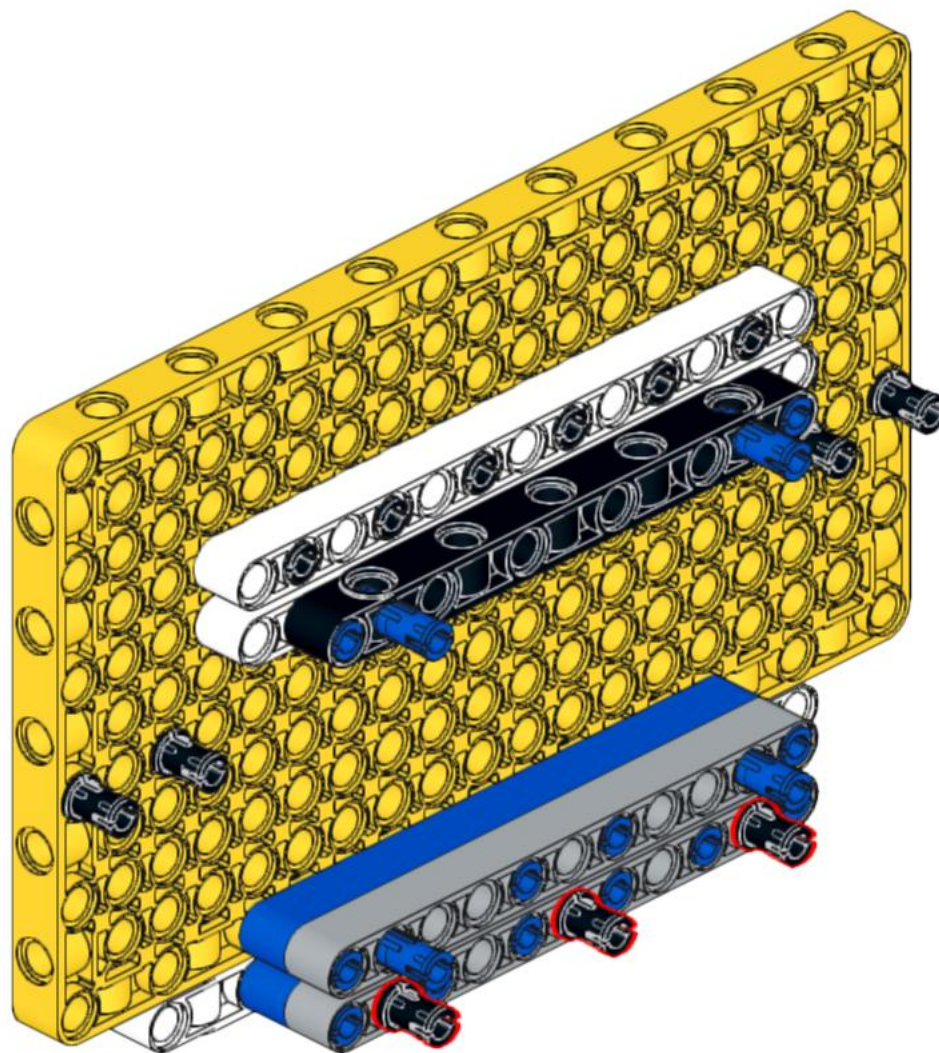
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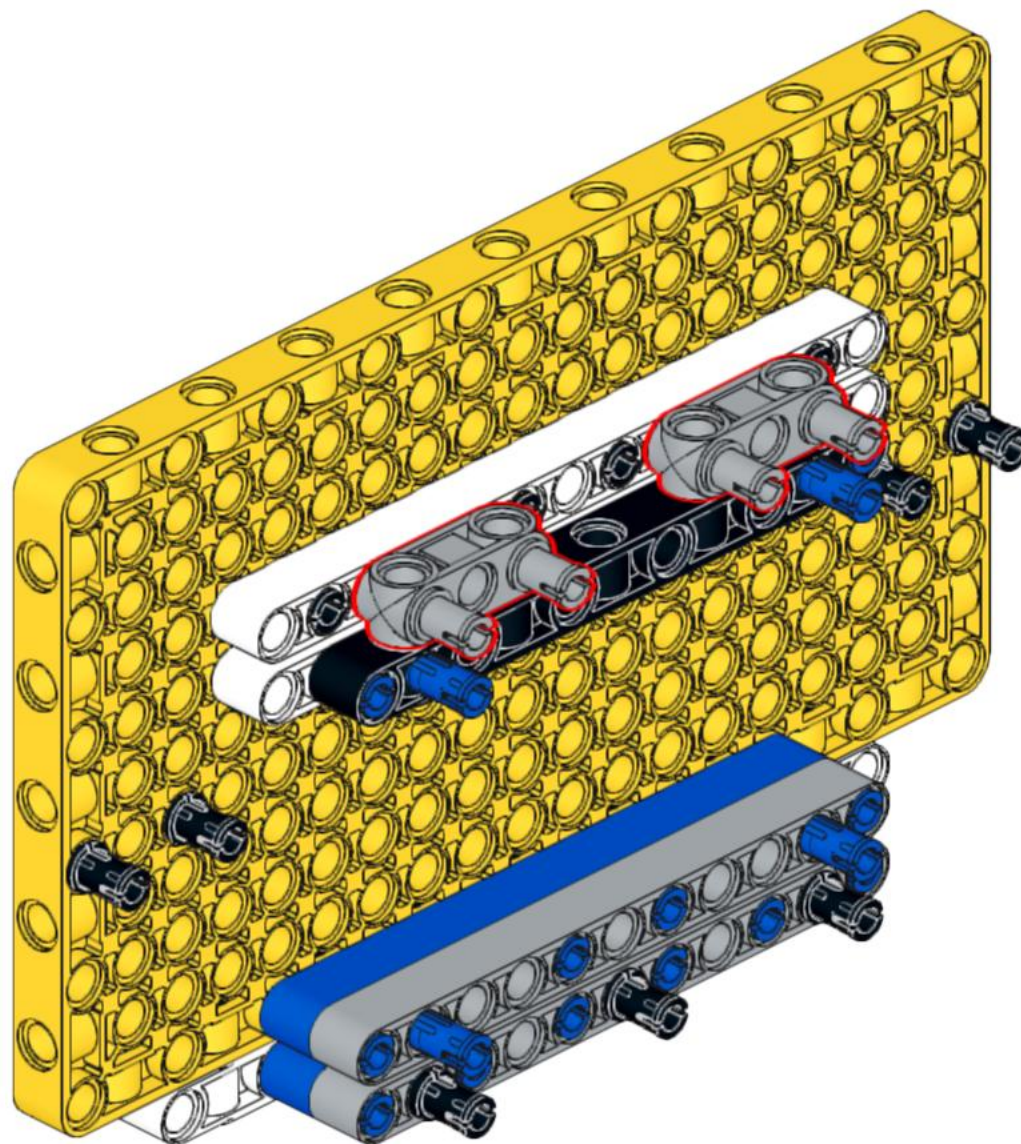
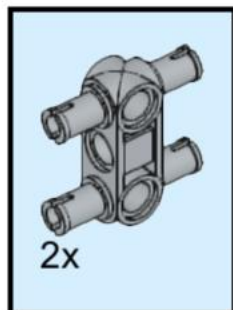
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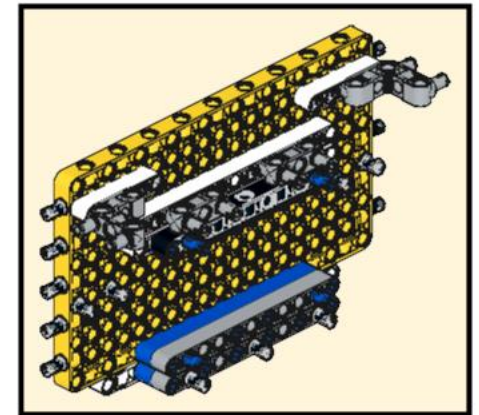
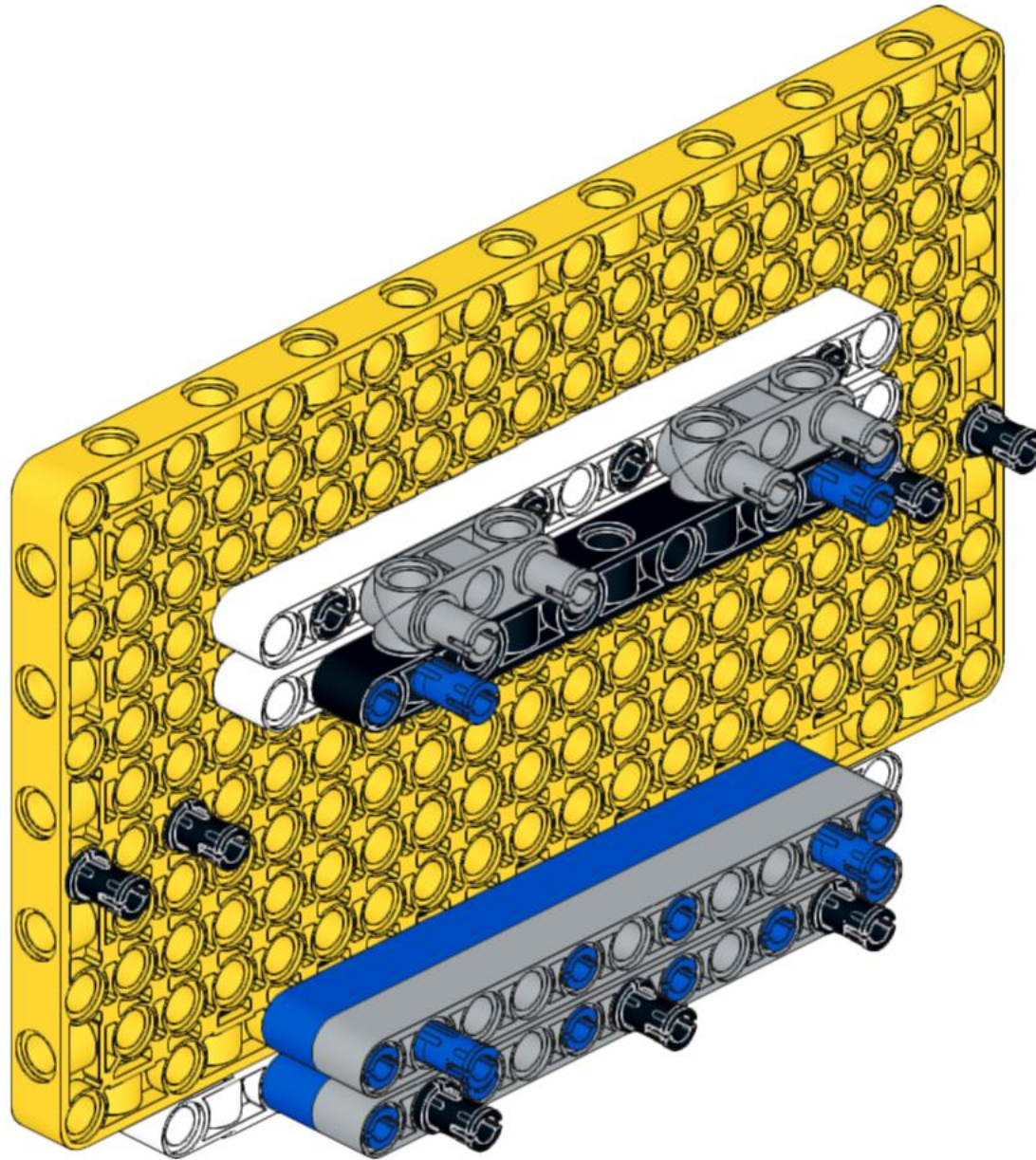
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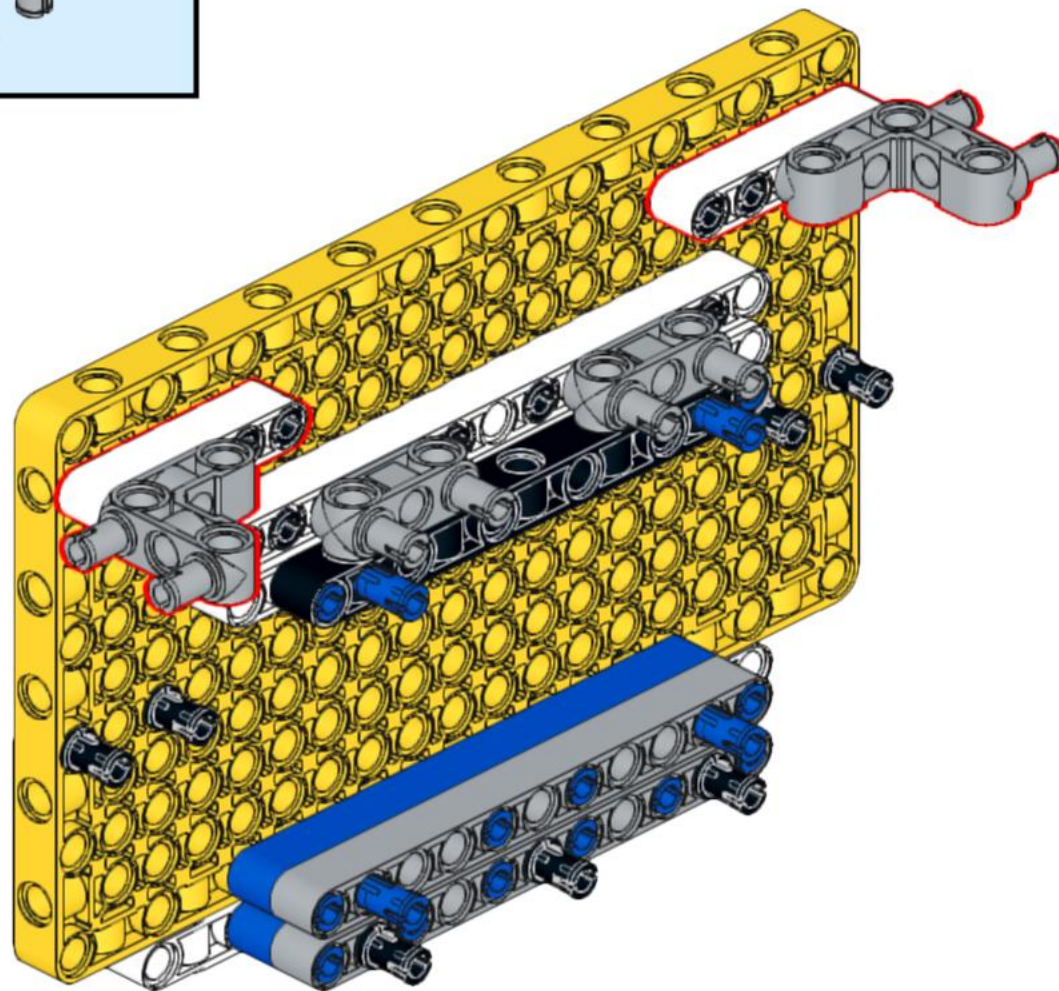
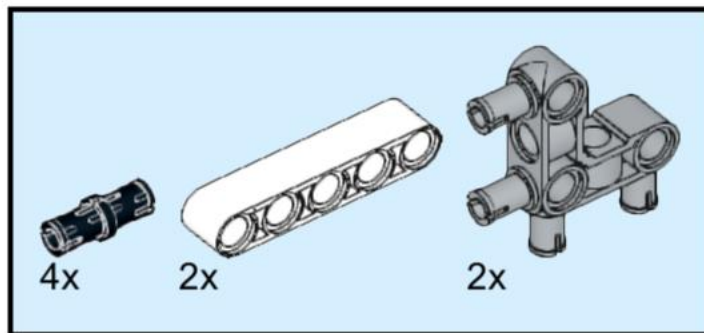
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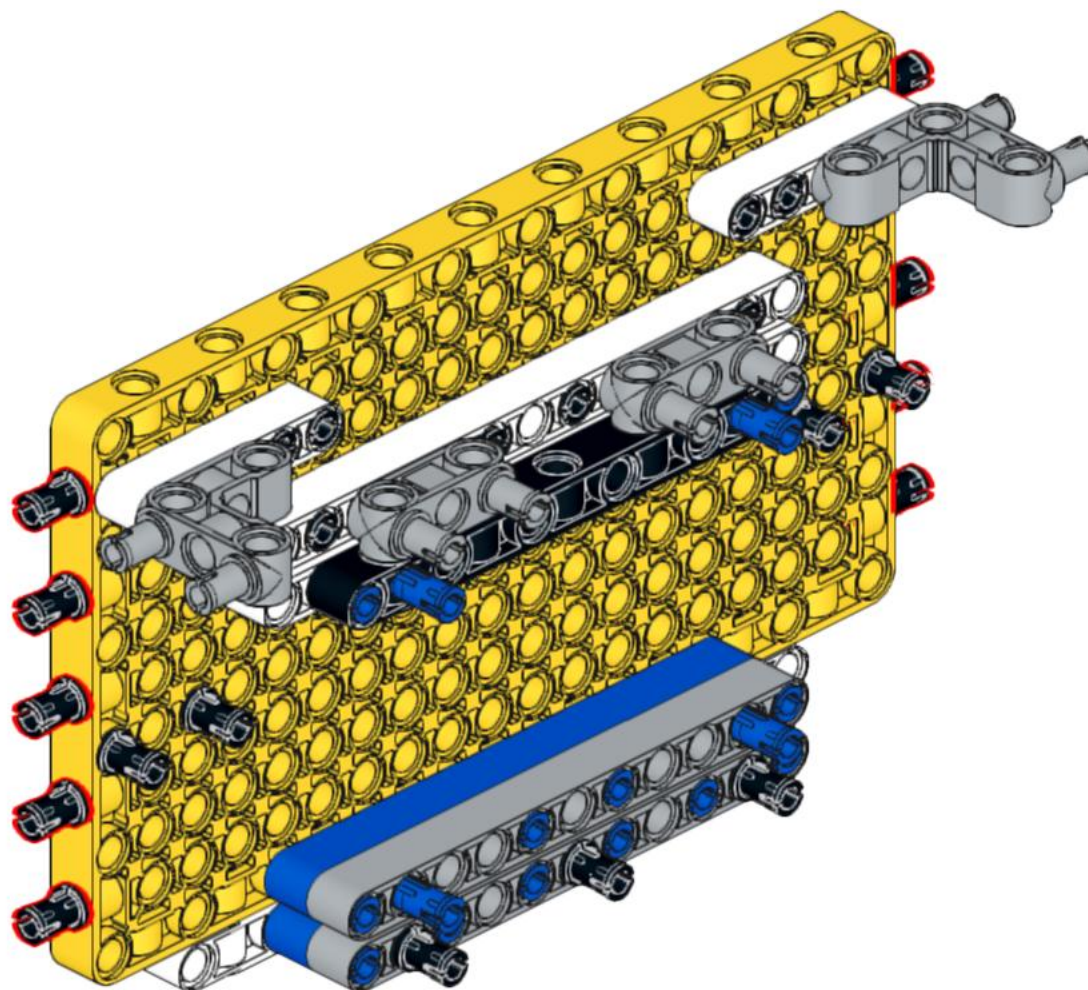
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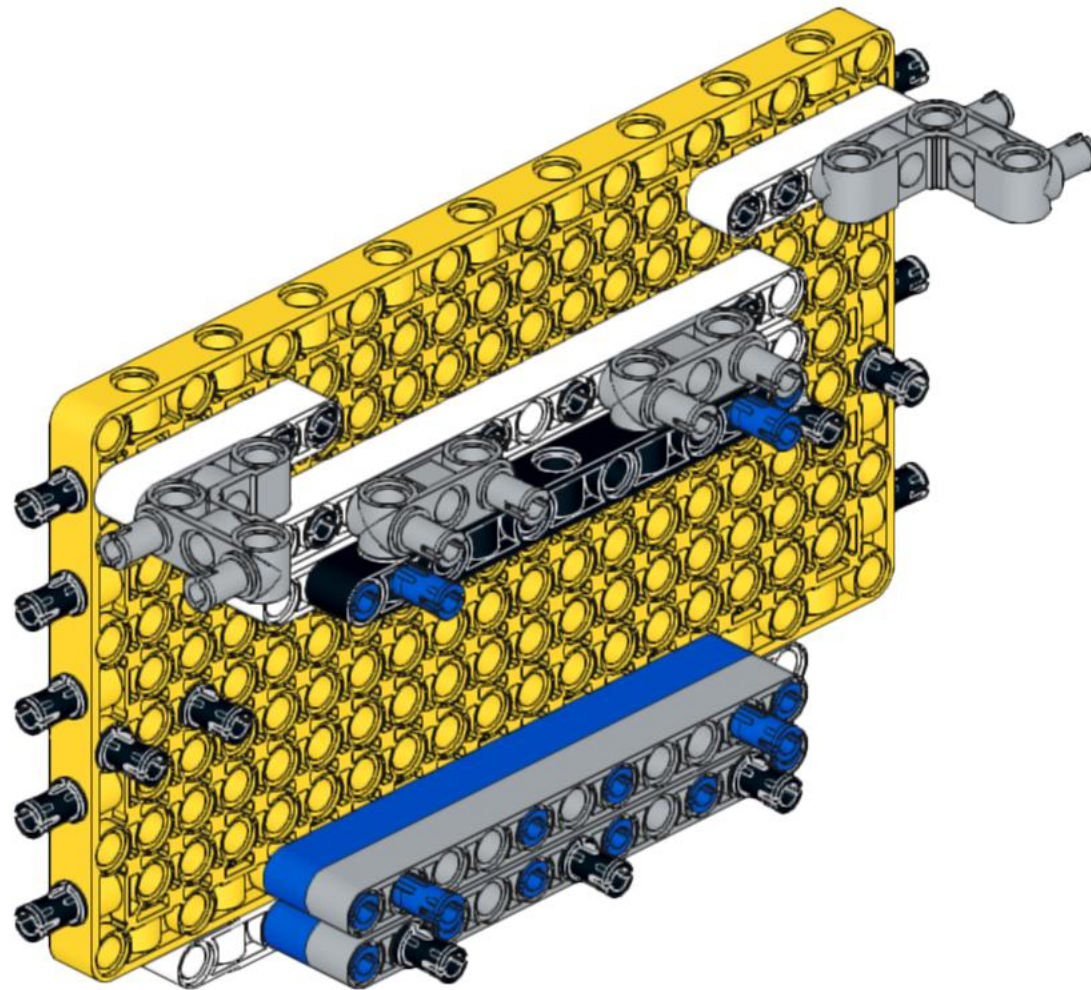
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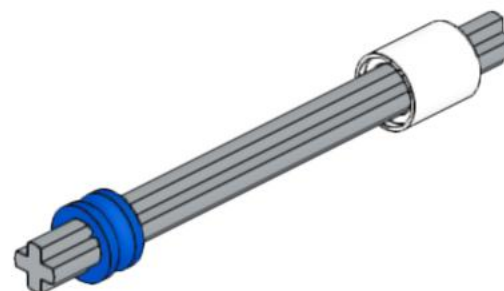
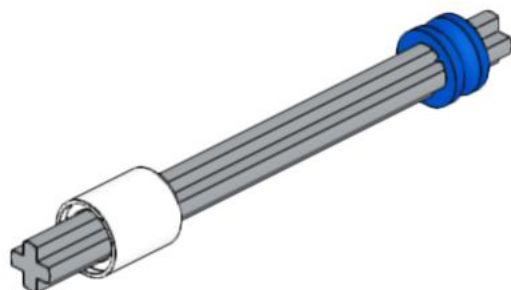
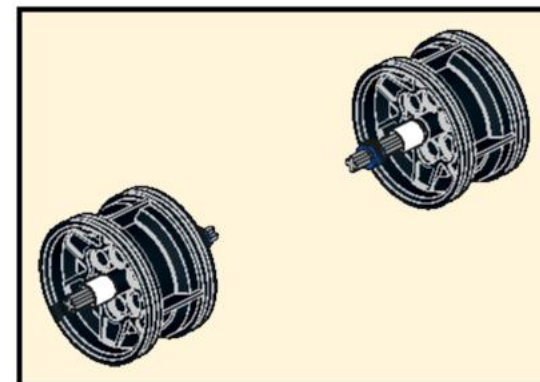
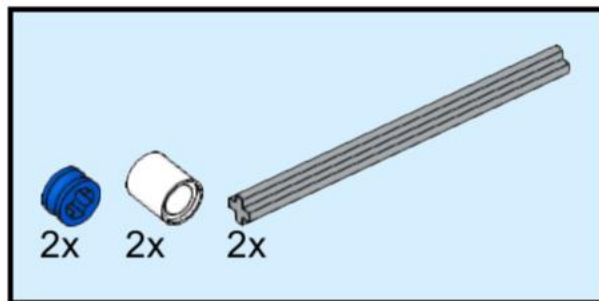
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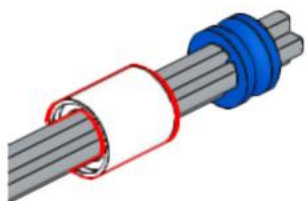
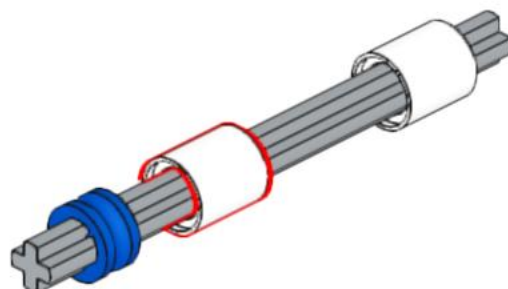
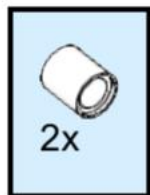
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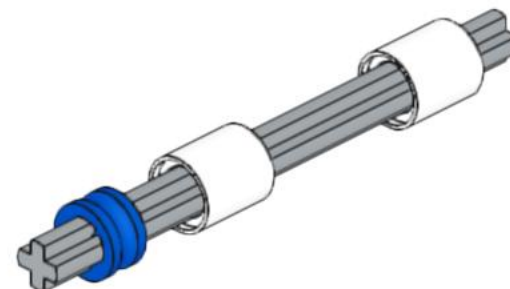
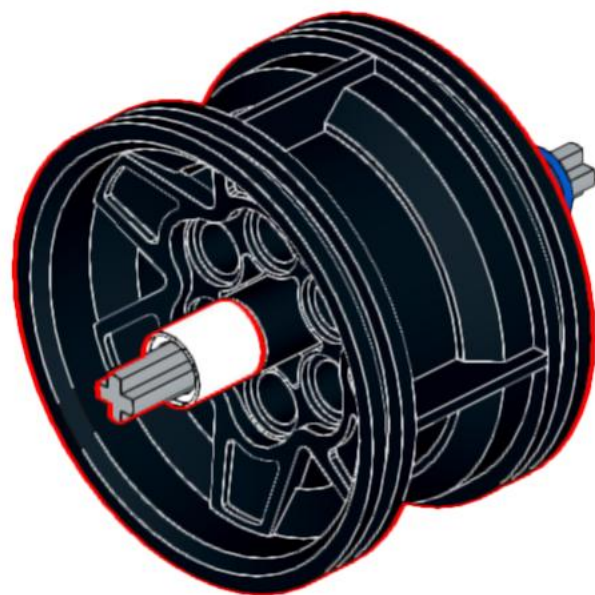
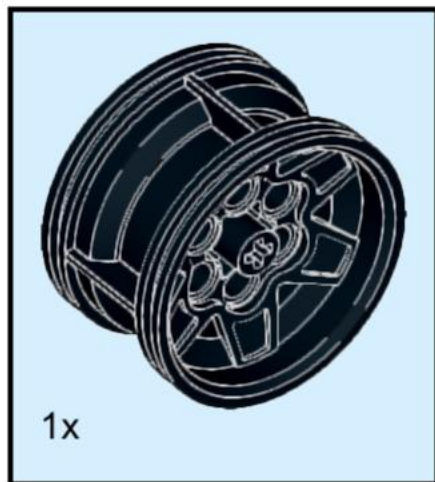
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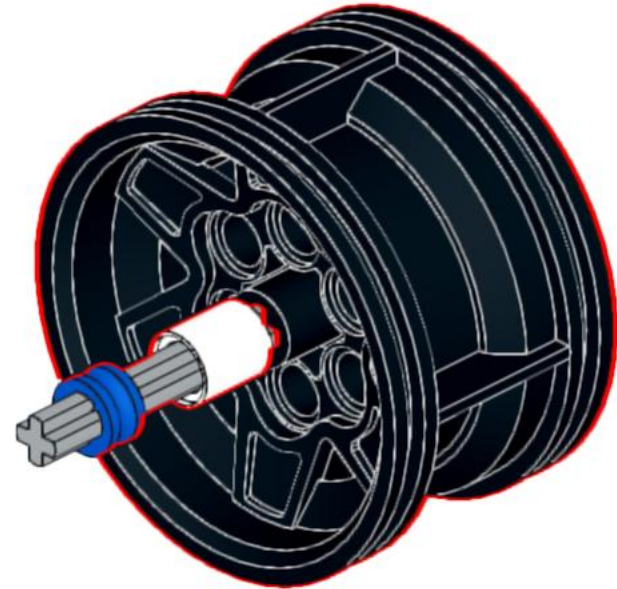
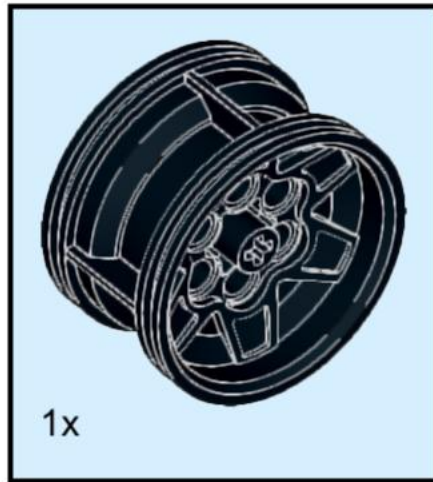
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49

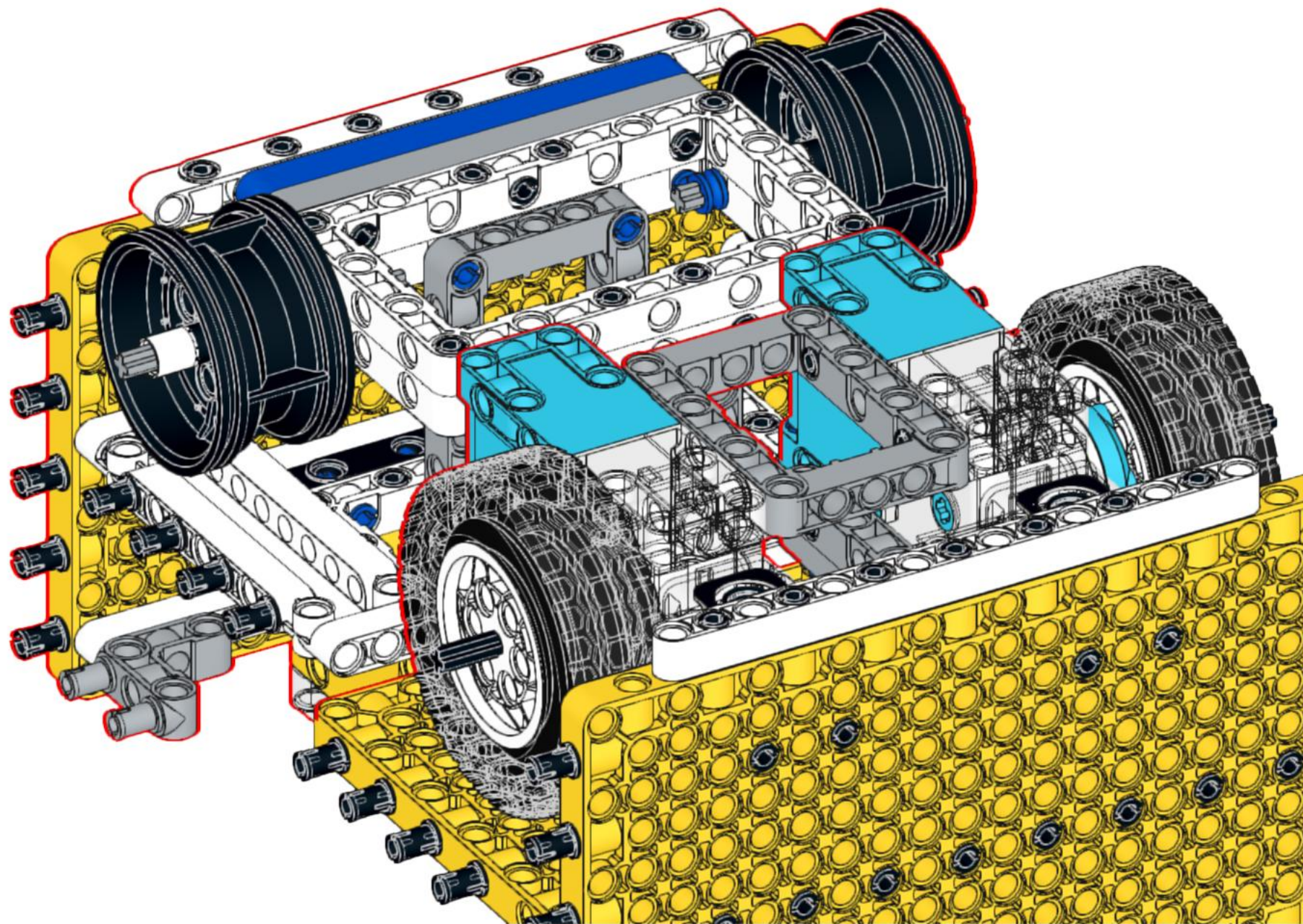


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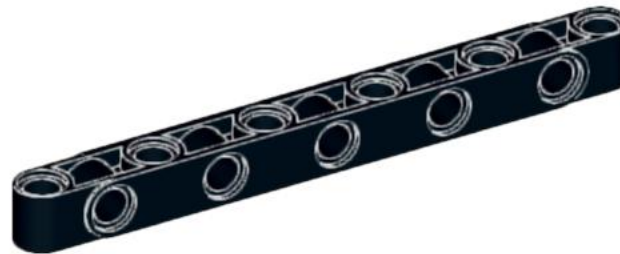
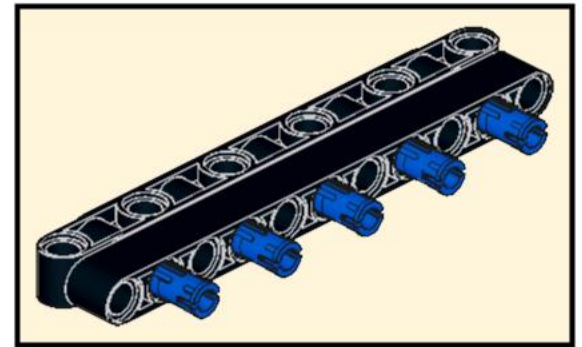
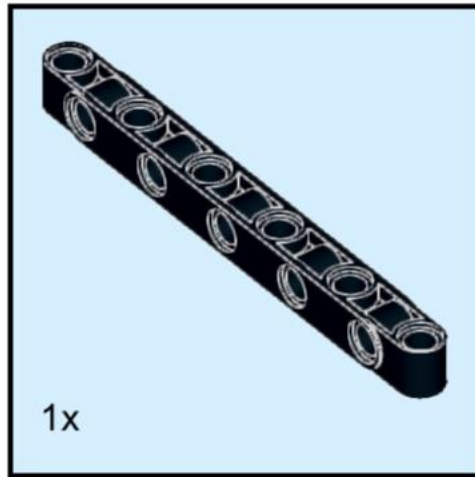


Hopefully it is obvious that these week assemblies will have be taken apart to install in the next step.

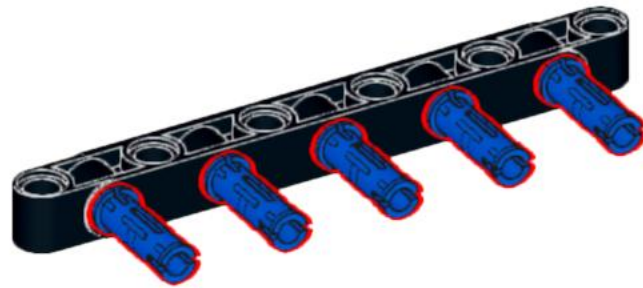
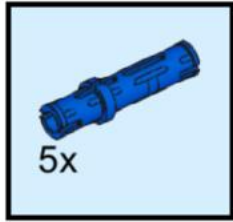
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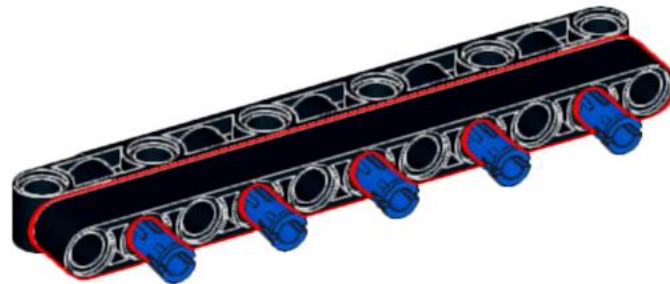
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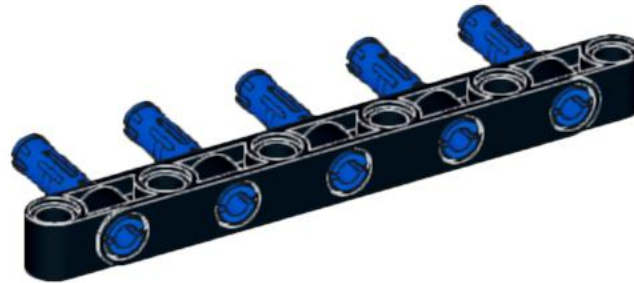
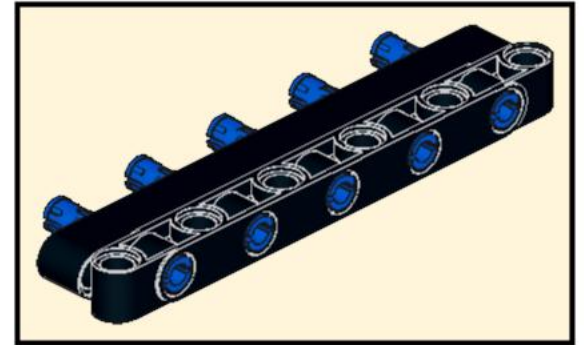
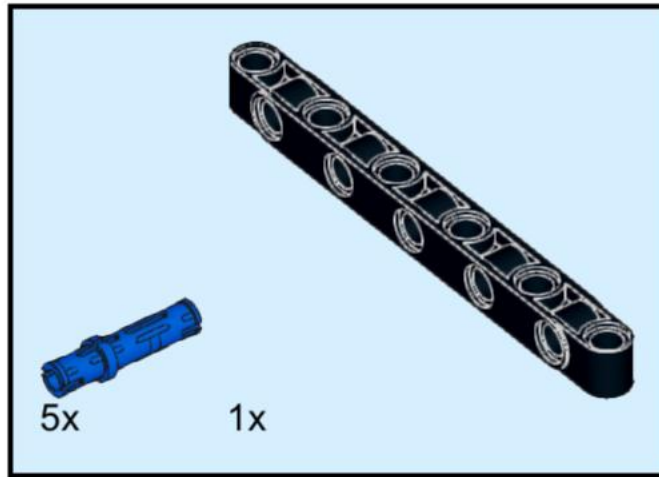
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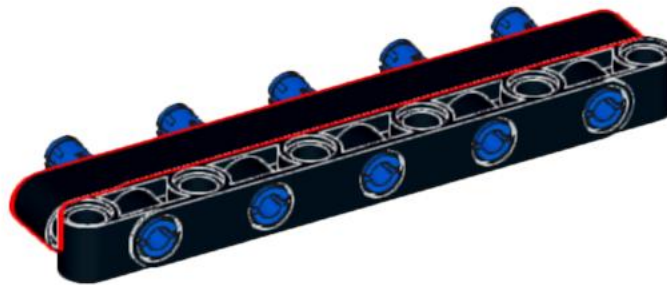
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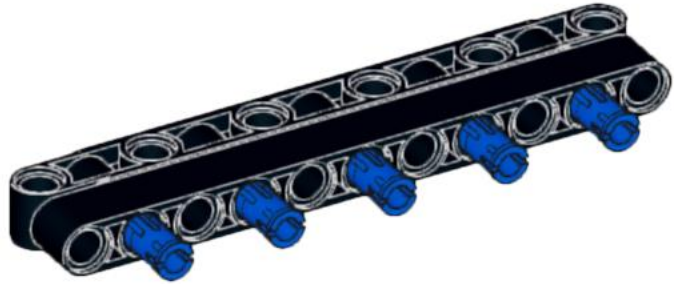
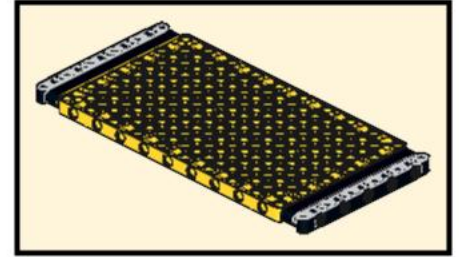
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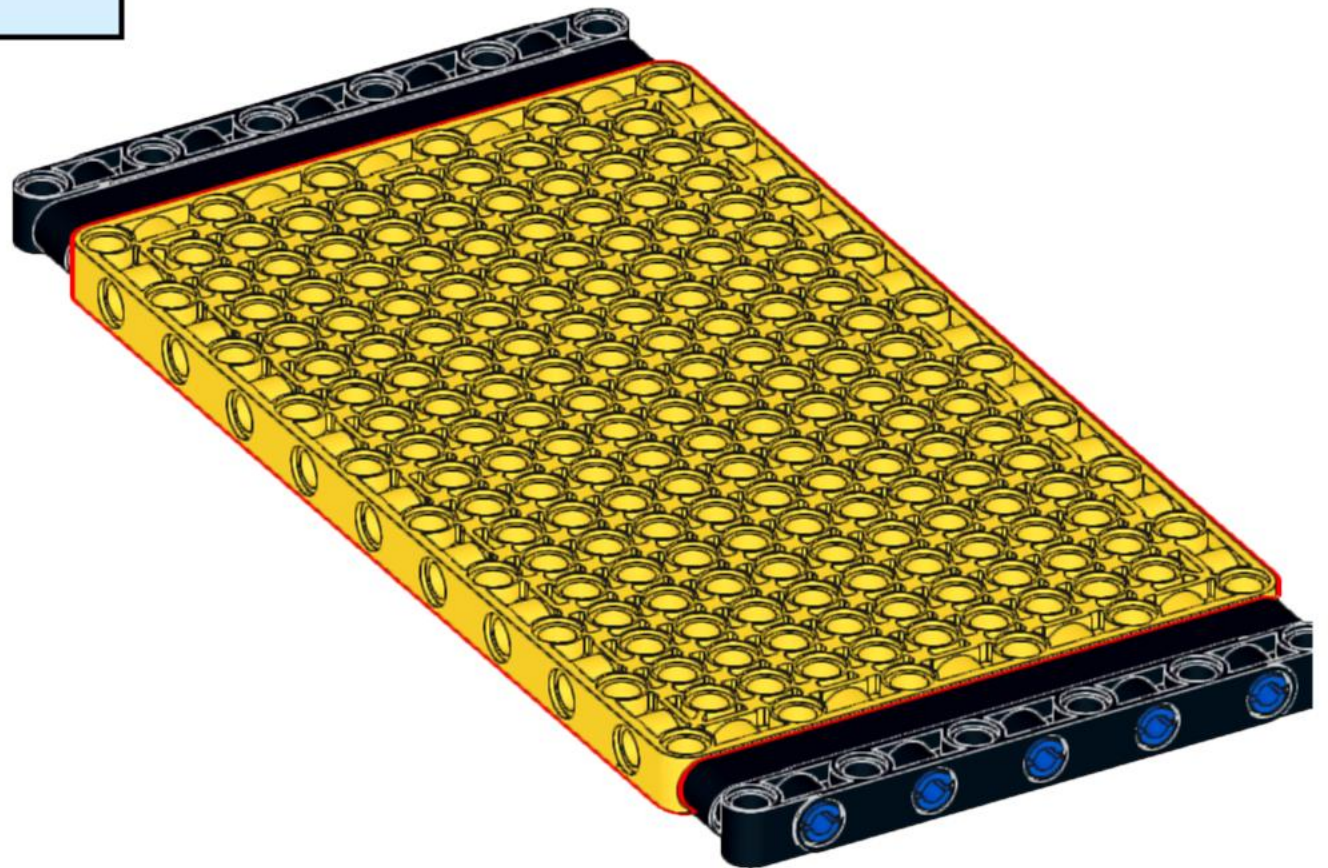
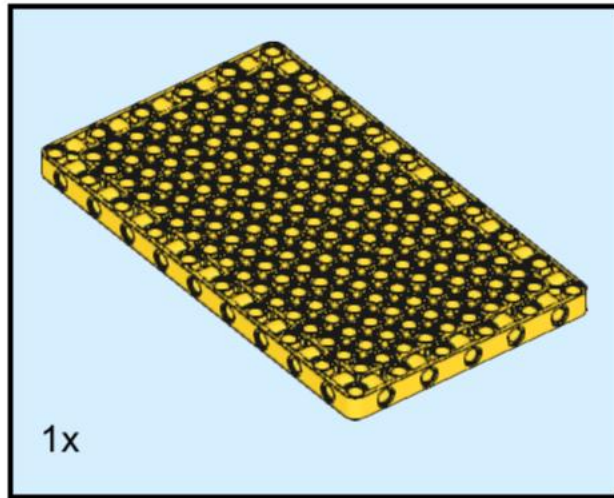
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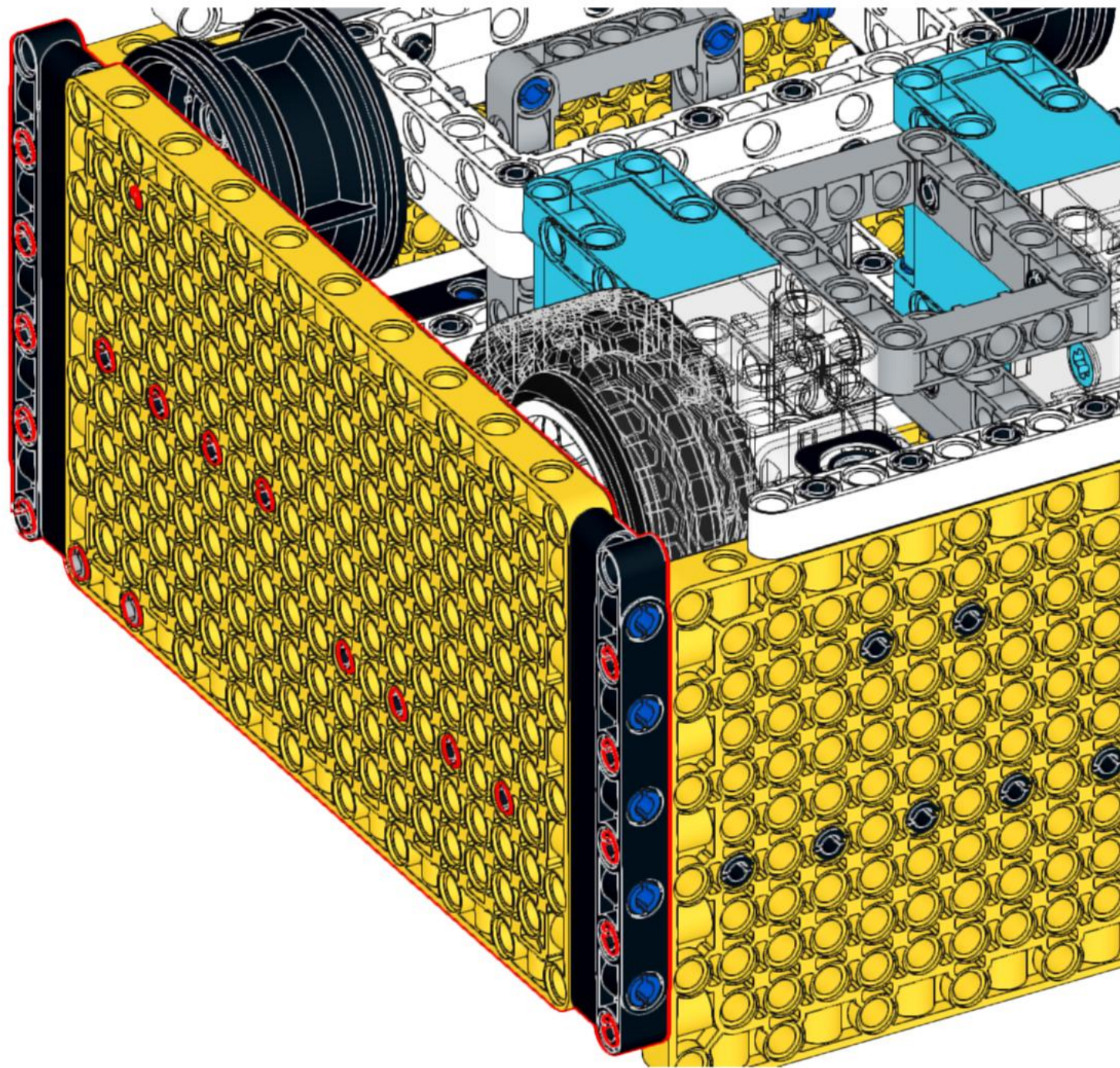
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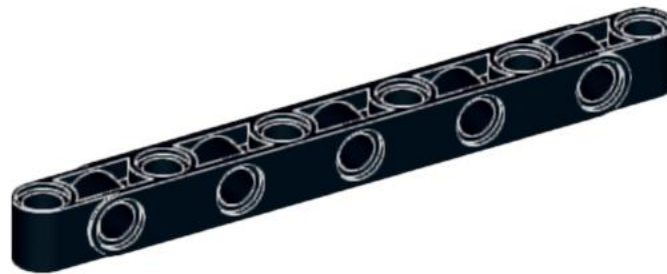
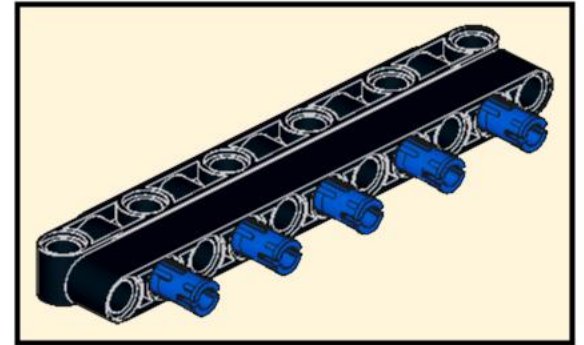
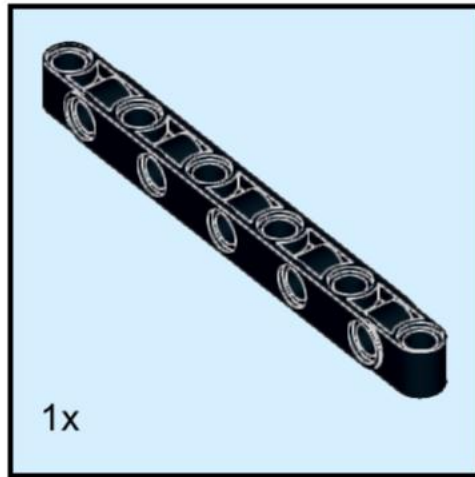
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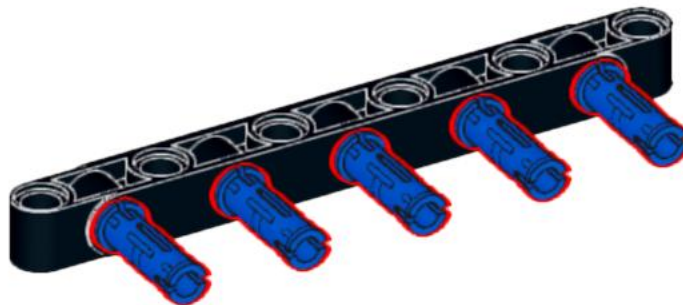
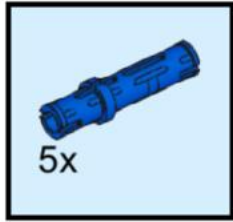
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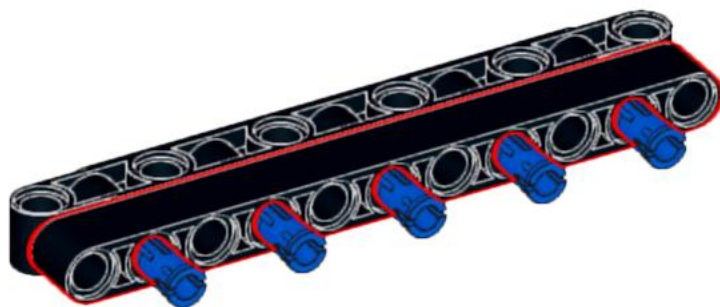
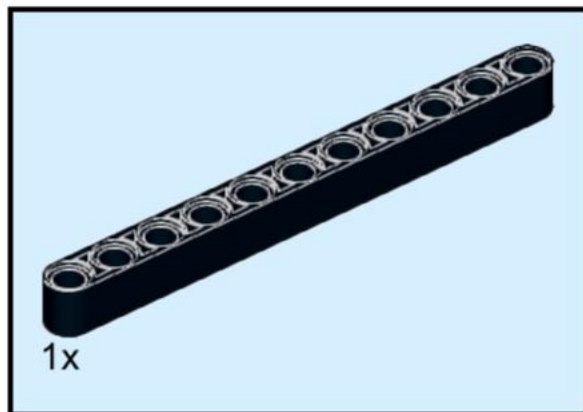
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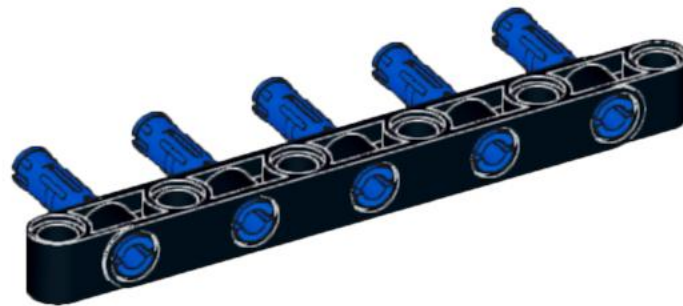
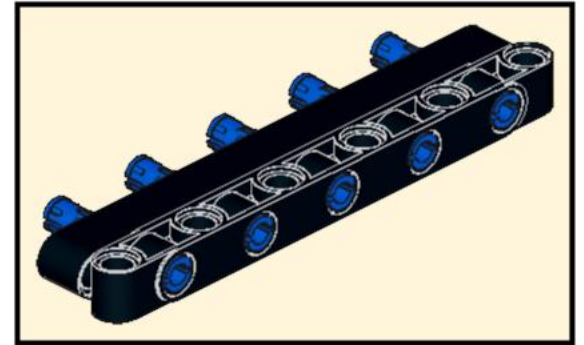
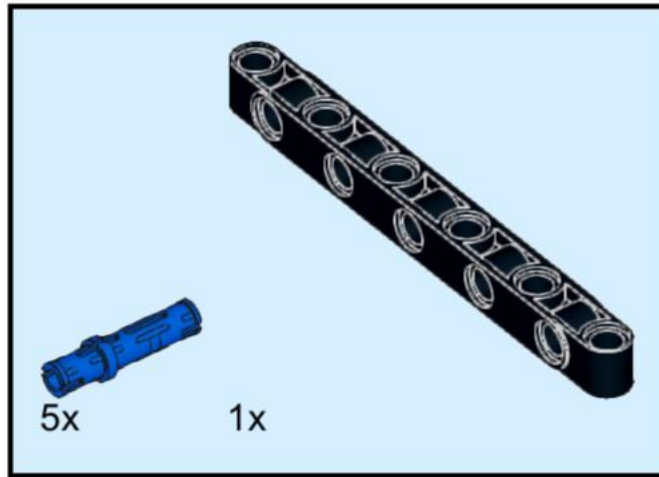
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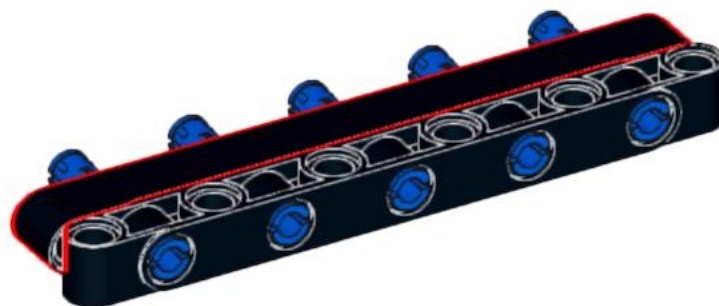
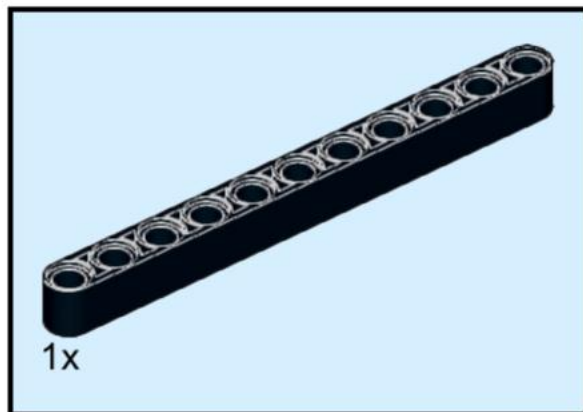
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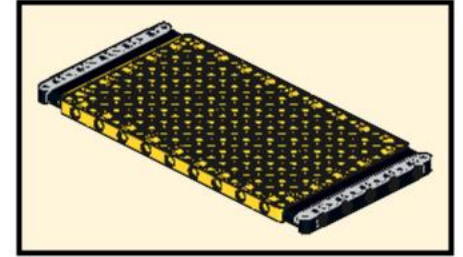
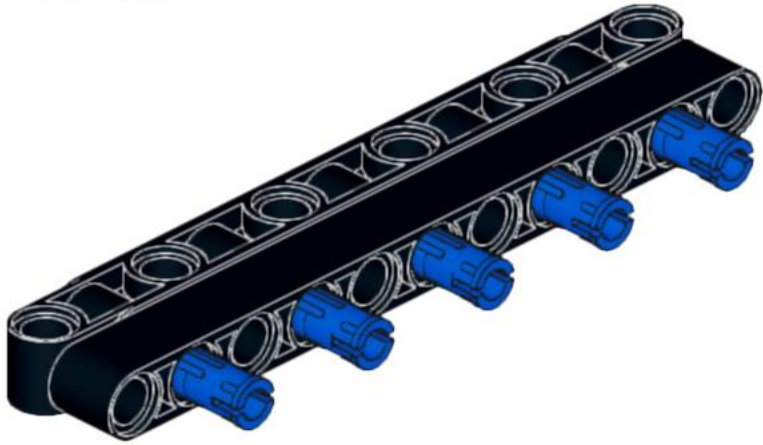
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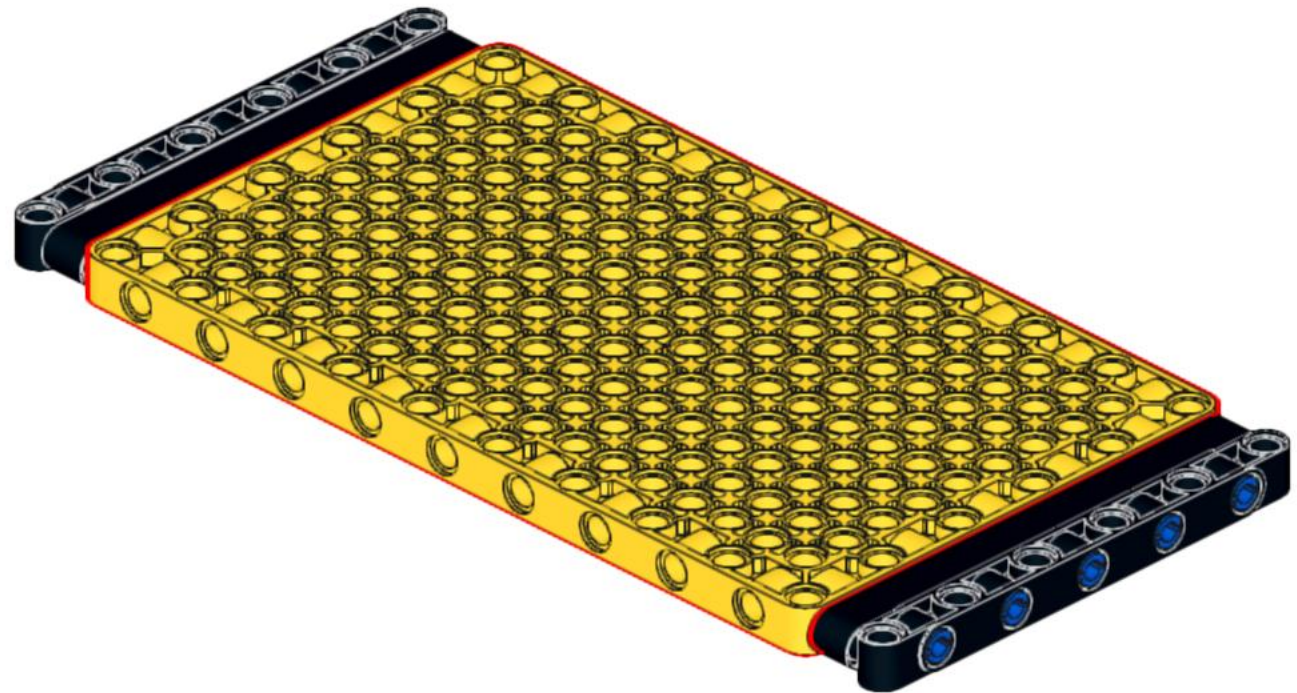
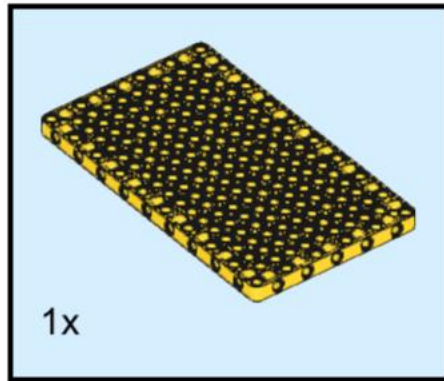
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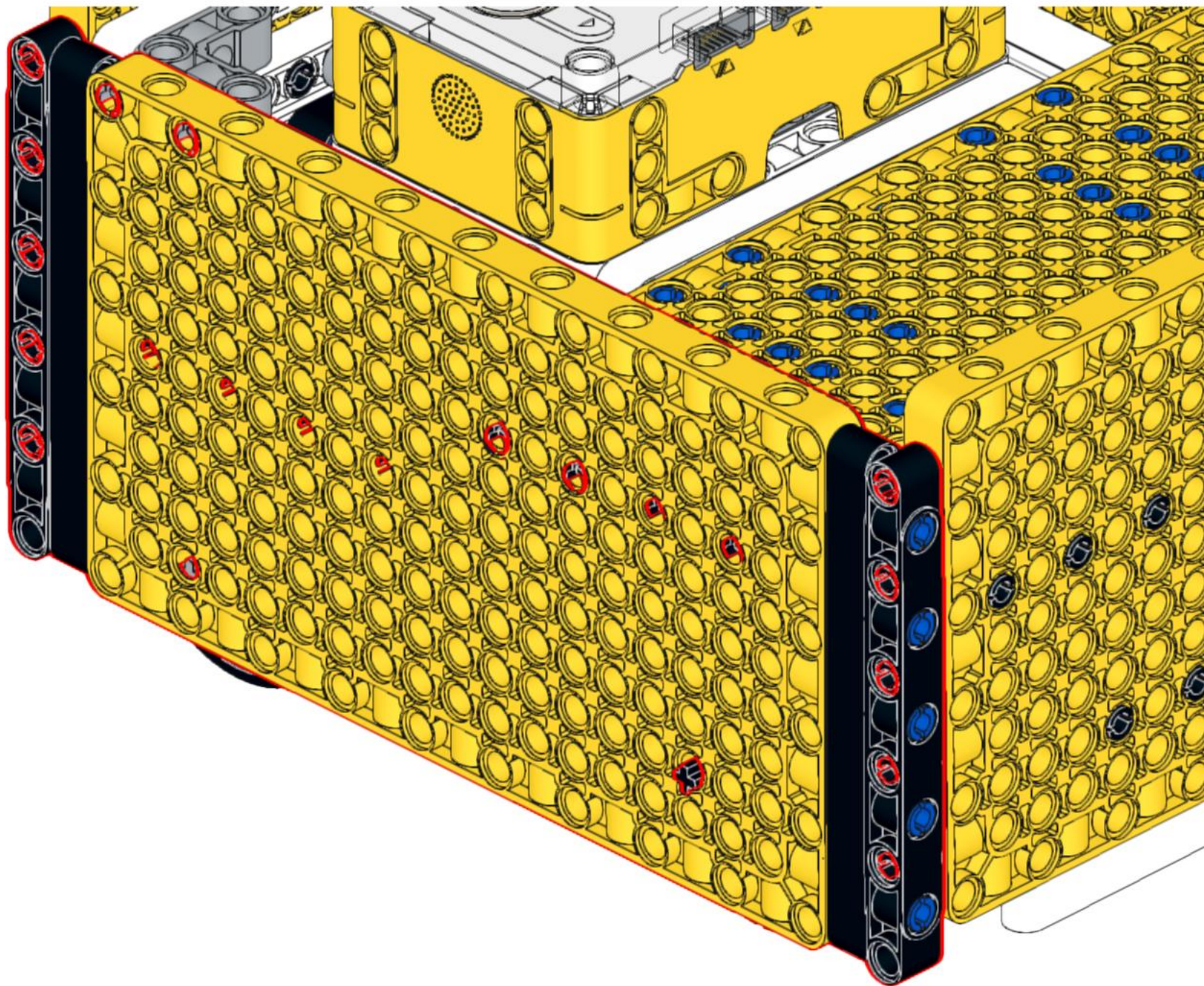
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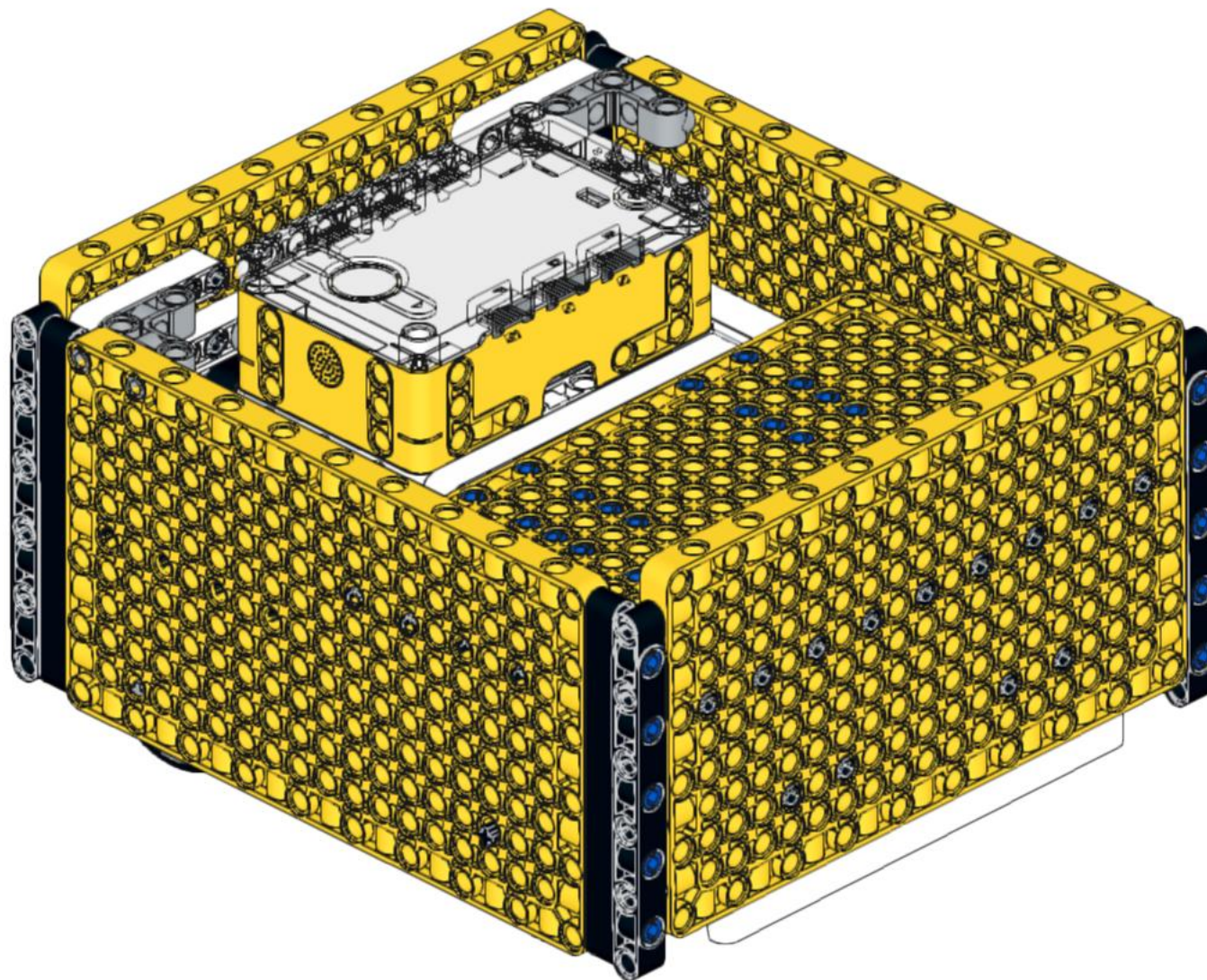
66



67



68



The core box robot is completed.

The next step is to install your motors. We recommend using the large Spike motors due to substantially better torque over the medium Spike motors.

There are a substantial number of ways you can install the motors based on how the team wants its attachments to work.